Internal Migration and Labor Market Adjustments in the Presence of Non-wage Compensation

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Abstract

In this paper, we argue that adjustments in non-wage compensation are empirically relevant and have important implications for understanding the effects of labor supply shocks. We examine the labor market impacts of internal migration in Brazil through a shift-share approach, which combines weather-induced migration with historical settlement patterns at each destination. Our findings indicate that increasing migration inflows lead to a reduction in formal employment while simultaneously increasing informality by a similar magnitude. Like previous studies, we observe a significant negative impact on earnings in the informal sector. Additionally, we provide evidence that the proportion of formal workers receiving non-wage benefits declines, underscoring that substantial adjustments take place in the formal sector, even in a context of high informality. We interpret our results within a framework that incorporates both formal and informal labor inputs, as well as non-wage benefits, and generates predictions closely aligned with our empirical findings.

Keywords: Internal migration, wages, employment, non-wage benefits.
JEL Codes: J2, J3, J61, O15.

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1 Introduction

Internal and international migration significantly influence development, demographics, and economic trends. In response, a vast body of literature has explored its impact on native populations, particularly concerning employment and wages. Borjas (2014), in his extensive work, highlights the costs of immigration for native workers who compete directly with immigrants. In contrast, many scholars argue that migration's effects are more nuanced (Card and Peri, 2016). For example, Card (2009) finds that immigration to the U.S. has only a minimal impact on native wages, while Ottaviano and Peri (2012) identifies minor positive wage effects for specific groups. This balance of perspectives emphasizes the complexity of immigration's economic consequences.

Canonical partial equilibrium models, which are characterized by perfect competition and the substitutability between native and migrant workers, suggest that wage adjustments can fully absorb labor market shocks when native workers are immobile. However, when wages are rigid, these models predict a decline in native employment (for an early example, see Altonji and Card, 1991). To reconcile the seemingly conflicting empirical findings, researchers have expanded these models to incorporate multiple outputs and technological change (Lewis, 2011; Dustmann and Glitz, 2015). They have also emphasized that different model specifications may capture distinct parameters (Dustmann et al., 2016).

In this paper, we argue that non-wage compensation adjustments are often overlooked in migration literature, yet they are empirically significant and can greatly influence the analysis of labor supply shocks caused by migration[1]. Specifically, we investigate the effects of internal migration in Brazil on the labor market outcomes for native workers within a context characterized by downward wage rigidity, where non-wage benefits play a crucial role in total compensation and labor informality is widespread. This environment allows us to examine how firms, when confronted with labor supply shocks due to migration, adjust both earnings and benefits in the formal sector, despite strict wage regulations and the prevalence of informal employment.

We motivate our analysis by developing a simple model based on the works of Harris and Todaro (1970) and Fields (1975), in which formal and informal employment coexist in destination regions. We extend their framework by incorporating non-wage benefits as a mechanism for adjusting the total compensation of formal sector workers, allowing employers to respond more flexibly to labor supply shocks. Additionally, our model assumes that production combines formal and informal labor with capital, treating formal and informal workers as substitutes rather than maintaining a strict segmentation between these sectors[2].

[1]For instance, Clemens (2021) suggests that considering non-wage adjustments may help clarify ongoing debates regarding minimum wage economics.

[2]This assumption is common in the literature, as discussed in the works of Ulyssea (2010, 2018) and

In the formal sector, firms must comply with minimum wage laws and frequently offer non-wage benefits, such as employer-provided health insurance, food, and transportation subsidies. In contrast, informal workers are not subject to these protections, and their earnings are fully flexible, following a standard competitive labor market. Our model predicts that, due to competition from incoming migrants, informal sector earnings are likely to decline. Furthermore, if the elasticity of substitution between formal and informal labor is high, migration inflows may also suppress the total compensation of formal workers. When formal wages cannot be adjusted downward because of minimum wage laws, firms may reduce non-wage benefits as a compensatory mechanism to absorb the shock.

We address the econometric concerns associated with the fact that migrants tend to move to areas with better labor market opportunities by taking advantage of a recent body of work that provides a clear framework for distinguishing sufficient conditions for identification and correctly computing standard errors (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2021; Jaeger et al., 2018; Adão et al., 2019). In particular, we combine two extensively used identification strategies into a shift-share instrument approach. First, we exploit exogenous rainfall and temperature shocks (or "shift") at the origin to predict the number of individuals leaving a municipality in the Brazilian Semiarid region. Then we leverage the history of the Semiarid as a large source of climate migrants and use the past settlement patterns (or "share") to allocate migration outflows to destination areas (Munshi, 2003; Boustan et al., 2010). The resulting predicted inflow of migrants is an instrument for observed migration. We use this shift-share IV design and annual data at the municipality-year level from 1996-1999 and 2001-2009 to identify the causal effects of immigration on labor market outcomes in the destination areas.

Following a weather-induced migration inflow, we find that earnings in informal and formal sectors decrease by 0.81% and 0.67% , respectively. Additionally, the share of native workers receiving food benefits declines by 0.24 percentage points, while transport benefits and health insurance decrease by 0.11 percentage points. Although overall employment remains unchanged, formal employment drops by 0.13 percentage points, with a corresponding rise in informality, highlighting a significant substitution between formal and informal labor. Those who lose formal jobs tend to transition into the informal sector, which operates without frictions like minimum wage laws. Contrary to existing literature, we observe negative impacts on formal sector earnings across various samples, except for low-education workers in markets with a high minimum wage bite. Research on internal migration in developing countries typically indicates that wage effects are concentrated in the informal sector or among low-education workers (Kleemans and Magruder, 2018; El-Badaoui et al., 2017). Our findings suggest that notable adjustments occur in the formal sector, even amidst high levels of informality, affecting earnings and the share of workers receiving benefits in most markets.

The heterogeneity analysis shows that the estimated effects on earnings are more pronounced in municipalities with a higher baseline level of informality, which aligns

Bosch and Esteban-Pretel (2012).

with our model predictions. We find that less educated native workers, who directly compete with migrants from Semiarid regions, experience more significant effects. These individuals are more likely to leave the formal sector and face larger earnings reductions in the informal sector compared to their more educated counterparts. Furthermore, since many low-education workers earn close to the minimum wage, the adjustments in benefit shares are expected to be more substantial. However, it is often the case that low-education workers who lose formal jobs are the ones receiving fewer benefits. This situation highlights the possibility of some 'positive selection': low-education workers who remain in formal employment tend to be more likely to receive benefits, which helps to alleviate the anticipated negative impact on benefit shares for this group.

Brazil offers a suitable environment for our investigation for three main reasons. First, over 3 million people in the Brazilian Semiarid, a historical source of climate migrants, left their hometowns during our sample period from 1996 to 2009. Second, conducting a within-country analysis reduces econometric concerns related to assigning migrants to specific skill groups (Dustmann et al., 2012). Third, more than 40 percent of workers are employed in the less regulated informal labor sector, where firms often do not comply with labor market laws, such as minimum wage and layoff regulations. In contrast, the formal sector employs the remaining workforce, where the minimum wage is typically above 70% of the median salary, and non-wage benefits are commonly offered. More than 31 million people, or 20% of registered workers, receive health insurance provided by their employers. After payroll expenses, this is the second-highest component of total labor costs (ANS, 2019). Additionally, 40% of these workers receive food subsidies, which cost employers approximately 57% of the minimum wage per worker.[3] To the extent that workers value non-wage benefits, changes in this margin of adjustment can have important welfare implications.

Our work is connected to a broad body of literature that explores how migration flows influence the labor market outcomes for native workers (see Borjas (2014) and Dustmann et al. (2016) for a review). While migration within countries is a more significant phenomenon—approximately 740 million internal migrants globally compared to an estimated three times fewer international migrants (UNDP, 2009; UN DESA, 2017)—most studies tend to focus on international immigration, particularly to high-income countries. These studies frequently concentrate on Mexican immigration to the United States (Borjas, 2003), as well as more recent immigration trends in Western Europe (Dustmann et al., 2012).

Some research indicates that native wages may be negatively affected by immigration (Borjas and Monras, 2017), whereas other studies suggest that the impact on native wages is minimal (Card, 2001) or even positive (Ottaviano et al., 2013; Foged and Peri, 2016; Azoulay et al., 2022). Dustmann et al. (2016) argue that these often contradictory estimates arise from different empirical specifications (sources of variation) and the fact that labor supply elasticity varies across different groups of natives. Additionally,

[3] Arbache and Ferreira (2001) based on various sources estimate the average cost of providing some job benefits in Brazil.

natives and immigrants do not compete in the labor market within the same education-experience cells.

An increasing number of studies are examining environmental and other economic shocks to analyze the causal impact of internal migration on labor markets in countries such as the United States, China, Indonesia, Thailand, and Brazil (Boustan et al., 2010; Hornbeck, 2012; Imbert et al., 2022; Kleemans and Magruder, 2018; El-Badaoui et al., 2017; Imbert and Ulyssea, 2024; Albert et al., 2021)[4]. In the context of developing countries, Kleemans and Magruder (2018) and El-Badaoui et al. (2017) closely relate to our work by showing that immigration leads to increased informality and lower earnings in informal sectors within destination regions. This finding aligns with the main predictions of Harris and Todaro (1970) and Fields (1975), which emphasize wage rigidity in the formal sector. Busso et al. (2021) provide several tests to assess the empirical relevance of the original Harris-Todaro formulation using Brazilian census data, highlighting the significance of incorporating additional factors such as urban informality and unemployment.

Conversely, Imbert and Ulyssea (2024) examine long-term changes using decennial data from 2000 to 2010. They find that internal migration reduces informality, as workers transition from informal to formal jobs. This is driven by an increase in the number of formal firms and job opportunities, without any changes in labor supply or unemployment. Their findings align with a model of firm dynamics and informality, suggesting that long-term labor market frictions, which constrain the expansion of formal demand, can be alleviated. Similarly, Albert et al. (2021) argue that frictions in the capital market and spatial limitations in the labor market hinder the reallocation of climate migrants from agriculture to manufacturing. This constraint limits the potential positive returns of migration, even over the long run. Although both studies focus on Brazil, Imbert et al. (2022) find that migrants integrate quickly into formal manufacturing firms, consistent with China's relatively flexible labor market.

While Imbert and Ulyssea (2024) concentrate on the long-term effects observed over a decade, our paper highlights the short-term negative impacts on employment, earnings, and non-wage benefits in the formal sector. Additionally, we provide unique insights into labor markets in developing countries by examining the interplay between informality, binding minimum wages, and non-wage benefits as an alternative margin of adjustment. Imbert and Ulyssea (2024) also show that, in the short run, workers reallocate from the formal to the informal sector, which is consistent with our findings.

Our contribution to the literature on the economics of migration is threefold. First, we demonstrate that firms systematically adjust non-wage benefits in response to labor supply shocks, and recognizing these adjustments is crucial for understanding the impact of migration on native workers. We find that, due to the high substitutability between formal and informal labor inputs, there are offsetting employment responses across formal and informal sectors. Second, we provide evidence on the effects of

[4]For a comprehensive review of the determinants of internal migration in the U.S., see Molloy et al. (2011), and for urban-rural internal movements, refer to Lagakos (2020).

internal migration on local labor markets in a large developing country. We show that these different adjustment patterns remain relevant even in the context of informality. Third, we contribute to a growing body of evidence indicating that migration serves as a significant coping mechanism against climate change, particularly for vulnerable populations in rural areas of developing countries (Skoufias et al., 2013; Assunção and Chein, 2016).

Non-wage benefits also play a significant role in compensation packages in developed countries. In the United States, employer-provided health insurance and other benefits make up about one-third of total compensation costs (Clemens et al., 2018) A study conducted in 2013 found that 74% of European firms offered non-base wage components, such as benefits and bonuses (Babecký et al., 2019) Research indicates that firms often modify these non-wage components in response to adverse economic conditions or as a strategy to counterbalance collective bargaining, especially when base wages are inflexible (Cardoso and Portugal, 2005; Babecký et al., 2012; Babecký et al., 2019) We contribute to this body of literature by demonstrating that non-wage benefits are an important margin of adjustment. Failing to consider them may lead to an underestimation of the effects of labor supply shocks on the formal sector caused by internal migration.

This paper is organized as follows. The next section presents background information on the Brazilian Semiarid region and local labor markets. Section 3 outlines a simple framework for interpreting our findings. Section 4 describes the data and empirical framework and reports the first-stage estimates that link the observed migration patterns to our predicted migration flows. Next, we present and analyze the main results on employment, earnings, and non-wage benefits in Section 5. We also study the sensitivity and heterogeneity of our main estimates. Finally, we interpret our main estimates in light of our simple model and conclude.

2 Background

In this section, we first describe the economic background and weather conditions in the Semiarid region, the functioning of local labor markets in Brazil, and a simple framework to contextualize our analysis. We then discuss the main data sources on labor market outcomes, migration flows, and weather, and present descriptive statistics.

2.1 Brazilian Semiarid

The Brazilian Semiarid region includes 960 municipalities spread across nine states, covering an area of approximately 976,000 km². This area is roughly equivalent to the combined territory of Germany and France. The Semiarid region accounts for 11 percent of Brazil's total land area and encompasses parts of nearly all Northeastern states, except for Maranhão, as well as the northern area of Minas Gerais. Notably, it does not include any state capitals. According to the official definition from the Ministry of National

Integration, a municipality qualifies as part of the Semiarid region if it meets at least one of the following three criteria: (i) an annual average precipitation of less than 800 mm between 1961 and 1990; (ii) an aridity index of up to 0.5 (the Thornthwaite Index, which measures humidity and aridity for a given area during that same period); or (iii) a drought risk greater than 60% (defined as the percentage of days experiencing hydric deficit, based on data from 1970 to 1990). Its average historical precipitation is about 780 mm, in contrast to approximately 1,600 mm for the rest of the country. The average temperature in this region is around 25°C. The rainy season typically occurs between November and April, with the highest levels of precipitation recorded after February, when the sowing season usually begins.

Municipalities are relatively small, with a median population of around 20,000, and their economies primarily rely on agriculture and cattle ranching within small subsistence farms (Rocha and Soares, 2015). Local economic activities are particularly vulnerable to weather shocks (Wang et al., 2004); some studies have indicated that agricultural production can decline by up to 80% during prolonged droughts (Kahn and Campus, 1992). Approximately 80% of the children in these municipalities live below the poverty line, and in 1996, the infant mortality rate was 31 per 1,000 births, compared to a national average of 25% and 15 per 1,000 births, respectively (Rocha and Soares, 2015). Additionally, over 80% of the adult population had less than eight years of schooling in 1991.

These poor socioeconomic indicators, especially during periods of extreme drought, have historically led to significant outflows of migrants, known as *retirantes*, from the Semiarid region to other parts of the country. During the 1960s and 1970s, net migration from areas outside the Northeast states (where most of the Semiarid region is located) reached 2.2 million and 3.0 million individuals (Carvalho and Garcia, 2002), resulting in net migration rates of -7.6% and -8.7%, respectively. From 1996 to 2010, approximately 3.0 million people left the Semiarid region in search of better living conditions elsewhere in the country. Appendix Figure C1 illustrates that these migrants tend to concentrate in certain states. Over the last four decades, São Paulo alone has received nearly 40% of the migrants arriving from the Semiarid region. In relative terms, the influx of migrants has led to an increase of over 2% in the population of the top 10 receiving states.

2.2 Labor Markets in Brazil

A common feature of labor markets in developing countries is the existence of a two-sector economy where the informal sector accounts for one to two-thirds of the GDP (see Perry et al. (2007) and Ulyssea (2020) for a review). In Brazil, more than 40% of the individuals work in the informal sector (those without registration or who do not contribute to social security), including most of the self-employed who are not protected through social security. When firms hire workers under a formal contract, they are subject to several legal obligations, such as paying minimum wages and complying with safety regulations. The registration also entitles workers to other benefits, such as a wage contract, which in Brazil prevents downward adjustment, working up to 44 hours a week, paid annual leave, paternity or maternity leave, retirement pension, unemployment insurance and severance payments (e.g. Gonzaga, 2003; Almeida and Carneiro, 2012; Meghir et al., 2015; Narita, 2020).

If firms do not comply with working regulations, they may be caught by the labor authorities and have to pay a fine. For example, a firm is fined about one minimum wage for each worker who is found to be unregistered, or the firm can be fined up to a third of a minimum wage per employee if it does not comply with mandatory contributions to the severance fund (Almeida and Carneiro, 2012).[5] It is well known that compliant (formal) firms are more visible to labor inspectors and therefore subject to more inspections, while informal firms tend to be smaller and are thus more difficult to catch. (Cardoso and Lage, 2006). There are additional anticipated costs for formal firms related to labor courts if a worker is terminated and chooses to file a lawsuit against the company. Judges tend to favor workers in nearly 80% of these cases (Corbi et al., 2022). This highlights a significant operational cost for the formal sector, especially for smaller businesses. Moreover, imperfect enforcement and costly regulations contribute to a high level of informality in the country.

There is significant overlap between the productivity distributions of the formal and informal sectors (Meghir et al., 2015), meaning that even at lower percentiles of the overall distribution, both sectors will be impacted by the arrival of migrants. In other words, both sectors include workers who can be considered close substitutes for the migrant workforce, leading to increased competition.

Non-wage compensation. Our empirical analysis focuses on three main fringe benefits identified in the data: private health insurance, food subsidies, and transport subsidies. In Brazil, these benefits gained popularity in the 1980s, with an increase in the frequency of food subsidies and employer-provided health insurance among private sector firms (Arbache, 1995). Data from PNAD surveys covering 1997-2009 show that 17% of the working-age native population receives food benefits, 18% receives transport benefits, and 9% obtains private health insurance through their employers.[6] Arbache and Ferreira (2001) estimates that benefits like food subsidies cost around 57% of one minimum wage, which is approximately 16% of average total compensation. Furthermore, data from the Brazilian Federal Health Agency (ANS, 2018) reveal that employer-provided health insurance averaged R$582 in 2018, representing 17% of total compensation for that year. These figures suggest that depending on how firms choose to allocate benefits within their workers' compensation packages, these expenses could account for more than 30% of total payroll costs. In the United States, benefits, including employer-provided health insurance, comprise about one-third of compensation costs (Clemens, 2021).

There are at least two reasons for the use of fringe benefits in workers' compensation. First, in Brazil, these benefits are not subject to payroll taxes, which helps to reduce

[5]The minimum wage is above 70% of Brazil's median wage.

[6]Among formal native workers, these percentages are 35%, 33%, and 18%, respectively.

overall labor costs. Second, labor laws are generally more flexible regarding the provision of benefits, making it easier to adjust these benefits compared to wages. Although regulations governing fringe benefits are typically less strict than those for wages, collective bargaining agreements (CBAs) sometimes include clauses related to these benefits. Notably, the third most common type of clause found in extended firm-level CBAs involves wage supplements, such as food subsidies. Furthermore, approximately 10% of all formal sector firms are covered by CBAs that include clauses on health plans or insurance. (Marinho, 2020).

The importance of adjustments in the non-wage compensation margin in Brazil, in comparison to other countries, raises an important question. Non-wage benefits should be more significant in countries with high minimum wages. In Brazil, the minimum wage accounts for 75% of the median wage, which is notably higher than in the United States (32%), Japan (44%), Mexico (46%), and several European countries, such as the UK (55%) and France (61%).[7]

Although transport subsidies have been a mandated benefit in Brazil since 1985, we regard this as a benefit that firms can adjust. This is supported by the fact that only 18% of formal sector native workers report receiving this benefit, indicating potential non-compliance by firms with labor regulations. Additionally, as transport benefits are classified as non-wage compensation, firms do not incur payroll taxes on them. Companies can also deduct the cost of the offered subsidy from their income tax base and operational expenses, which ultimately lowers their net revenues and affects various corporate and payroll taxes[8]. This situation creates incentives for firms to offer transport benefits and further motivation to adjust these benefits — either by providing better transportation options or by increasing the cash equivalent of the benefit.

3 A Simple Theory

This section describes a simple model assuming perfectly competitive labor markets to guide our analysis. We assume that migrants and natives are perfect substitutes and investigate the consequences of a migration shock that shifts the aggregate labor supply to the right. Then, we introduce intersectoral linkages where formal and informal workers are inputs that can be combined in a production function.

We begin with a standard competitive market model, in which the wage is determined, such as when labor supply equals labor demand and migration negatively affects wages. In the extreme case of an inelastic supply, migration does not affect the employment of natives, and the entire migrant workforce is absorbed. On the other hand, with an upward-sloping labor supply, the wage decline makes jobs less attractive for some native workers, and native employment falls.

Of course, this benchmark vastly simplifies how the labor market works. Downward

[7] Brazil (Source: PNAD Continua 2015). Data for OECD countries in 2019 (Source: OECD.Stat, https://stats.oecd.org/Index.aspx?DataSetCode=MIN2AVE

[8] The income tax due cannot be reduced by more than 10%.

wage rigidities are often present in reality due to minimum wage laws and collective bargaining agreements. In this case, migration shocks can be accommodated by job losses or lower labor costs, for example, reducing non-wage benefits (McKenzie, 1980; Clemens, 2021). In developing countries' labor markets, excess workers can also be accommodated in the competitive informal sector.

This traditional view of informality considers that the formal and informal sectors are segmented, which masks important intersectoral linkages, in particular, on the production side (see Ulyssea (2010, 2018) and Bosch and Esteban-Pretel (2012)). To address such context, we develop an extension of a model with informality in which the regulated sector is subject to minimum wage laws. This is guided by previous work by Kleemans and Magruder (2018) that builds on Harris and Todaro (1970) among others[9], and we modify their model in two ways.[10] First, we add non-wage benefits as a source of adjustment of total compensation for formal workers in the presence of binding minimum wages. Second, we assume that production combines formal and informal labor besides capital. We follow the literature and allow formal and informal inputs to substitute in production.[11] Firms comply with minimum wages by paying formal workers at least $\underline{W}$, where this is binding. They can also offer non-wage benefits frequently observed in the data (e.g., employer-provided health insurance and food). On the other hand, informal workers are not entitled to minimum wages or non-wage benefits, and their wages are completely flexible as in a standard competitive setting.

Production is given by

$$Y = K^{1-\alpha} L^{\alpha},\ 0 < \alpha < 1 \tag{1}$$

where labor combines two inputs, formal and informal, that can be substituted with elasticity $1/(1-\nu)$ as described below,

$$L = (\theta_f L_f^{\nu} + \theta_i L_i^{\nu})^{1/\nu} \tag{2}$$

with and θ_f and θ_i denoting the share parameters of formal and informal labor inputs, respectively, and $0 < \nu < 1$ reflecting heterogeneities across formal and informal labor.

Marginal products for workers of each type determine wages (or total compensation). For formal sector workers, if we assume that firms can freely adjust non-wage compensation but can only reduce wages up to the binding minimum, then total compensation equates to their marginal product, such that

$$W_f = \underline{W} + B = \frac{\alpha \theta_f K^{1-\alpha}}{L_f^{1-\nu}} (\theta_f L_f^{\nu} + \theta_i L_i^{\nu})^{(\alpha-\nu)/\nu} \tag{3}$$

[9] e.g. Fields (1975) and Mazumdar (1976). In the same spirit, Busso et al. (2021) considers two urban sectors - formal and informal - and unemployment.

[10] Kleemans and Magruder (2018) also follow Card and Lemieux (2001) and Borjas (2003)'s model to address the differences in the impacts of immigration by skill, which in their context (Indonesia) is crucial given that the poor natives were less educated than the migrant workforce.

[11] In Brazil and Mexico, respectively, 40% and 44% of informal labor are employed in formal firms (Ulyssea, 2018).

where W is the wage ($W \geq \underline{W}$) and B is the cost of non-wage benefits to employers.

For informal workers, wages are freely determined. This implies that wages are

$$W_i = \frac{\alpha \theta_i K^{1-\alpha}}{L_i^{1-\nu}} (\theta_f L_f^\nu + \theta_i L_i^\nu)^{(\alpha-\nu)/\nu} \tag{4}$$

Now, consider that the immigration labor supply shock affects informal workers, L_i.[12] This is consistent with migrants and informal workers being more similar to low-education natives. Our model leads to the following predictions derived in Appendix A:

(i) $\frac{\partial W_i}{\partial L_i} < 0$

(ii) $\alpha - \nu < 0 \Rightarrow \frac{\partial W_f}{\partial L_i} < 0$

These statements show a basic intuition for the effects of internal migration of workers who are similar to native workers in the informal sector. First, informal wages are expected to decline due to competition with similar incoming migrants. Second, formal-sector employers can decrease wages, employment, and/or non-wage benefits. However, minimum wages constrain them in some situations, unlike the informal sector. Thus, we should see more significant decreases in employment and non-wage benefits where the minimum wage is more binding. In such settings, the effect on non-wage compensation depends on the degree of substitution between formal and informal inputs. If the elasticity of substitution is higher than the labor share, then an inflow of informal workers lowers non-wage compensation of formal workers.

Impacts of migration on formal sector earnings, employment, and non-wage benefits are likely heterogeneous across regions depending on the share of workers receiving non-wage benefits. In situations where the minimum wage is more binding, we would expect larger decreases in employment and smaller decreases in non-wage benefits when workers were not receiving non-wage benefits to begin with. In contrast, we may see a larger decrease in non-wage benefits and smaller decreases in employment where a large fraction of workers were receiving non-wage benefits beforehand.

The model also accounts for variations in outcomes across submarkets, in markets characterized by higher baseline informality (high θ_i), the negative impacts on wages or total compensation are shown to intensify as the share of informality rises (see Appendix A).

4 Data and Empirical Strategy

This section begins by listing the main data sources used in our analysis and showing some descriptive statistics. Then, we describe the empirical framework and report first-stage estimates that link observed migration patterns to our predicted migration flows.

[12] This can be thought of as a supply shock in the number of low-education workers. Given that the informal sector is more intensive in low-skilled labor, it should be the one affected directly.

Migration We use data from three waves of the Brazilian Census (1991, 2000, and 2010), provided by the *Instituto Brasileiro de Geografia e Estatística (IBGE)*, to create key variables for our study.[13] First, we analyze Census responses regarding the municipality of origin and year of migration to calculate the annual migration outflow from each municipality in the Semiarid region and the inflow to each destination (all areas outside the Semiarid region) during the 1996-2010 period. Second, we use data from the 1991 Census to create a "past settlement," measure, which associates the proportion of migrants from each Semiarid municipality residing in each destination. Further details on the structure of our yearly migration dataset can be found in Appendix C.

Weather shocks Weather data were obtained from the Climatic Research Unit at the University of East Anglia (Harris et al., 2020). The CRU Time Series offers global monthly gridded data on precipitation and temperature at a resolution of $0.5° \times 0.5°$, which corresponds to approximately 56 km^2 at the equator. To develop municipality-level monthly measures of precipitation and temperature, we calculate a weighted average of the raw data from the four grid nodes surrounding each municipality. The weights are determined by the inverse of the distance to the grid's centroid.[14] We define a municipality's rainfall and temperature shocks as deviations from its historical averages.[15]

Labor outcomes Our primary data source for the outcome variables is the *Pesquisa Nacional por Amostra de Domicílios (PNAD)*, a national household survey conducted by the *IBGE*, the agency responsible for the Census. The PNAD survey is carried out annually, with the exception of Census years. Therefore, our data covers the periods from 1996 to 1999 and from 2001 to 2009. Interviews are conducted in 808 municipalities across all 27 Brazilian states, addressing various dimensions such as education, labor, income, fertility, and household infrastructure. On average, approximately 300,000 individuals are interviewed in each round of the survey.

Two key features of the PNAD (National Household Sample Survey) warrant further examination regarding the suitability of its data for our analysis. First, it is important to note that less than 20% of all Brazilian municipalities are represented in the survey. Nonetheless, these municipalities account for nearly 80% of the destinations chosen by migrants from the Semiarid region and comprise 65% of the employed population in Brazil. This concentration indicates that, despite its limited geographic coverage, the survey effectively captures a significant portion of migration trends and labor market dynamics. Second, the PNAD is not designed to be a representative survey at the municipality level. To validate its use as our primary source for labor outcome

[13] As some municipalities were divided into new ones during the 1990s, we aggregate our data based on the original municipal boundaries as they existed in 1991 (referred to as "minimum comparable areas" or MCA) to avoid potential misclassification related to migration status or municipality of origin. Throughout this paper, we use the terms municipality and MCA interchangeably.

[14] This methodology is similar to the one employed by Rocha and Soares (2015).

[15] For a detailed description and discussion about this measure, see Appendix G.

measures, we present simple descriptive statistics and empirical distributions of the main variables from both the PNAD and the Census in Appendix B. Appendix Table B1 contains these sociodemographic statistics. Furthermore, Figures B1 and B2 compare municipality-level employment rates and log earnings in both formal and informal sectors from the 2009 PNAD with corresponding data from the 2010 Census for the same municipalities. This comparison demonstrates significant overlap in both the formal and informal labor markets. Unfortunately, the Census lacks information on non-wage benefits. Additionally, Figures B3 and B4 present the correlation between employment rates and earnings in both sectors, utilizing PNAD data from 1996, 2001, and 2009, which align more closely with the census samples from 1991, 2000, and 2010. We argue that the outcomes from these datasets exhibit similar patterns.

Our primary outcomes focus on earnings and employment. We examine whether individuals are employed in the registered formal sector, the informal sector, or are self-employed. Additionally, we gather information on non-wage compensation. The survey includes questions about whether individuals received any form of payment or assistance for food or transportation expenses, as well as whether their employer provides health insurance. Unfortunately, data on the intensive margin of non-wage compensation is not available.

We restrict our sample to individuals aged 18 to 65 who have lived in the municipality for 10 years or more, referring to them as natives. We consider as destinations all municipalities from the PNAD that are not located in the Semiarid region, to reduce concerns about spatial correlation in weather shocks. We combined 13 years of individual survey data and calculated averages at the municipality-year level. Since our primary analyses use first differences, the final sample of destinations includes 684 unique municipalities and 8,190 municipality-year observations, averaging a population of 2,152,950 individuals.

Table 1 describes municipality-level data for the origin (Panel A) and the destination (Panel B) municipalities. Semiarid areas show lower levels of rainfall, slightly higher temperatures, and are less populated than destination municipalities. On average, 1.0 p.p. of Semiarid's population leaves every year, resulting in an average increase of 0.28 p.p. of the labor force in the destination.

Table 2 provides descriptive statistics for native individuals in the destination municipalities. In our sample, 63% of individuals are employed - with 35% having a formal job, while 28% are informal workers. The unemployment rate is 12%, and 25% of individuals are not in the labor force. The average monthly earning is R$ 849.39, with the formal sector having a substantially higher average (R$ 1,017.43) than the informal sector (R$ 634.58).[16] Among the working-age native individuals, 17% receive financial help to cover expenses with food, 18% for transport, and 9% for health expenditures.[17]

Finally, Table 3 compares migrants to all natives and natives by education level. We consider low-education individuals with up to seven years of schooling. Migrants have

[16]Earnings are measured in R$ (2012).

[17]Less than 1% of informal and self-employed workers receive any form of non-wage compensation.

a higher share of less-educated workers, are younger, more single, less white, and earn significantly less than natives. Compared to less-educated natives, they earn slightly more, have a similar likelihood of working part-time, and are somewhat more likely to be employed in the formal sector when compared to low-education natives. On the other hand, highly educated natives are more likely to work in the formal sector and have considerably higher pay.

Appendix Table C1 shows that top occupations for migrants (e.g., typically bricklayer for men, domestic worker for women) also represent a large share of the occupations among low-education natives, but not for the high-education. Also, the same five industries that concentrate almost 70% of working migrants employ a similar share of low-education workers (see Appendix Table C2). This characterization is consistent with greater substitutability between migrants and less educated natives in the labor market.

4.1 Empirical Strategy

Here, we first describe the empirical framework that allows us to (i) isolate the observed variation in migration induced by exogenous weather shocks and (ii) the migration flows into destination municipalities determined by past settlements. Next, we discuss and present supportive evidence on the validity of this shift-share instrument approach based on insights from the recent econometric literature that analyzes its formal structure.

We specify a model for the changes in the labor market outcomes of native individuals as a function of internal migration flows. Specifically, we assume that

$$\Delta y_{dt} = \alpha + \beta m_{dt} + \gamma X_d + \psi_t + \epsilon_{dt} \tag{5}$$

where y_{dt} is a vector of labor outcomes at destination municipality d in year t, m_{dt} is the destination migrant inflow from the Semiarid region, X_d are destination-level controls, ψ_t absorb time fixed effects and ϵ_{dt} is the error term. The main challenge in identifying β is that the observed migration, m_{dt}, reflects an equilibrium between the demand and supply of migrants. Additionally, the error term, ϵ_{dt}, may encompass unobserved characteristics that are correlated with migration inflows. Specifically, migrants may select a particular destination municipality in response to demand shocks that result in higher wages or better job prospects. We calculate the first difference in the outcome variables to control for time-invariant unobserved characteristics that could correlate with migrant inflows. However, this method does not account for time-varying confounders, which could bias OLS estimates.

To tackle the problem of endogeneity, we employ a two-step procedure to create an instrumental variable for the number of migrants entering a destination. First, we predict m_{ot}, the migration outflow rate, which is defined as the number of migrants leaving municipality o divided by the population recorded in the 1991 Census, for the year t. This prediction is made using weather shocks from the previous year:

$$m_{ot} = \alpha + \beta' Z_{ot-1} + \phi_o + \delta_t + \varepsilon_{ot} \tag{6}$$

where Z represents a vector of rainfall and temperature shocks experienced in the origin municipality o during the previous year. The terms ϕ_o and δ_t account for fixed effects for origin municipality and year, respectively. Additionally, ε_{ot} denotes a random error term. To estimate the number of migrants leaving their hometowns each year, we multiply this predicted migration rate by the municipality's population as reported in the 1991 Census:

$$\widehat{M}_{ot} = \widehat{m}_{ot} \times P_o \tag{7}$$

In the second step, we use the past settlement pattern to allocate migrants from the origin o to municipality d across various destination areas. Specifically, we define the proportion of migrants from origin o who settled in destination d during 1991. To address concerns about the persistence of migrant flows, as discussed by Jaeger et al. (2018), we fix our measure of past settlements in 1991 for the duration of our sample period.

$$s_{od} = \frac{M_{od}}{\sum_d M_{od}} \tag{8}$$

allowing us to define our shift-share instrumental variable (SSIV) as

$$\widehat{m}_{dt} = \sum_{o=1}^{O} \frac{s_{od} \times \widehat{M}_{ot}}{N_d} \tag{9}$$

where N_d is the total native population at d in 1991. Thus, our instrument $\widehat{m}_{dt}$ can be thought of as a combination of exogenous shocks or 'shifts' $\widehat{M}_{ot}$ (weather-driven outflows) with 'shares' ($s_{od} \geq 0$) or 'ethnic enclaves' as in Card (2001).[18]

The validity of the shift-share instrument approach relies on assumptions about the shocks, exposure shares, or both, as discussed by recent literature, which analyzes its formal structure. Goldsmith-Pinkham et al. (2020) demonstrate that a sufficient condition for consistency of the estimator is the strict exogeneity of the shares. Alternatively, Borusyak et al. (2021) show how one can instead use the exogenous variation of shocks for identification by estimating a transformed but equivalent regression - at the origin level in our setup - where shocks are used directly as an instrument.

Based on these insights, we leverage origin-level weather shocks[19] for identification and define the reduced-form relationship that associates labor market outcomes and the predicted migrant flow at the destination as

$$\Delta y_{dt} = \alpha + \beta \widehat{m}_{dt} + \gamma X_d + \psi_t + \epsilon_{dt} \tag{10}$$

[18]In appendix G we discuss this approach in more detail.

[19]Figure 1 illustrates the variation in weather shocks that are used for identification.

Building on the work of Borusyak et al. (2021), we compute a weighted average version of equation 10 at the origin level, using the exposure shares s_{od} as weights. This yields the transformed reduced form relationship.

$$\bar{y}_{ot} = \alpha + \beta \widehat{M}_{ot} + \bar{\varepsilon}_{ot} \tag{10'}$$

In Appendix H, we provide a detailed derivation of the transformation performed. As discussed in Borusyak et al. (2021), the consistency of our shift-share approach is based on two conditions:

Assumption 1 (*Quasi-random shock assignment*): $\mathbb{E}[Z_{\tilde{o}}|\bar{e}, s] = \mu$ for all $\tilde{o}$.

Assumption 2 (*Many uncorrelated shocks*): $\mathbb{E}[\sum_o s_o^2] \to 0$ and $Cov[Z_{\tilde{o}}, Z_{\tilde{o}'}|\bar{e}, s] = 0$ for all $\tilde{o}, \tilde{o}'$.

where $\tilde{o} = (o, t)$, $\bar{e} = \{\bar{e}_{\tilde{o}}\}_{\tilde{o}}$, $s_o = \sum_d s_{od}$ and $s = \{s_o\}_o$. As in Borusyak et al. (2021), $\bar{e}_{ot}$ corresponds to the error term from equation 5 computed at the level of shocks (e.g., municipality of origin). Assumption 1 guarantees that our shift-share IV is valid when weather shocks are as-good-as-randomly assigned, which comes from standard natural shocks arguments. Given identification, Assumption 2 gives us consistency when the number of observed shocks is large and when shocks are mutually uncorrelated given the unobservables and s_o. In the Appendix Table H1, we show that the effective sample size[20] is sufficiently large, reassuring us that exposure concentration is not a relevant issue in our setting. Also, in Appendix Tables I2-I4, we present evidence that the shocks we are using can be treated as uncorrelated, which supports the validity of Assumption 2 in our setting.

One additional advantage of using origin-level shocks is related to hypothesis testing. Adão et al. (2019) demonstrate that conventional inference in shift-share regressions is often invalid because observations with similar exposure shares are likely to have correlated residuals. This correlation can result in the overrejection of null hypotheses. Borusyak et al. (2021) argue that by employing the shock-level relationship instead of the destination-level, researchers can achieve standard errors that converge to those obtained through the correction procedure proposed by Adão et al. (2019).

4.2 Weather-induced Migration

We start our exploration of the first-stage results by estimating variations of specification 6, and we report the estimates in Table 4. All regressions account for temperature shocks, the logarithm of the total population from the previous census, and both time and municipality fixed effects. In columns (2) to (6), we include a flexible trend by interacting time dummies with characteristics from 1991, such as age and the proportions

[20] According to Borusyak et al. (2021), the effective sample size is the inverse of the Herfindahl index of concentration of migrants: $H = \frac{1}{\sum_o s_o^2}$.

of individuals with high school and college education. Columns (3) to (6) feature up to three lags, contemporaneous data, and one lead of rainfall and temperature shocks. For brevity, we have omitted the mostly insignificant coefficients related to temperature shocks in Table 4. Standard errors are clustered at the grid level to account for the fact that municipalities within the same grid are likely to experience similar shocks. [21]

As anticipated, rainfall shocks from the previous year are negatively correlated with migration outflows, indicating that residents of the Semiarid region tend to migrate away during droughts. The coefficient estimates are remarkably stable across specifications, and adding additional lags does not alter the baseline results. Significantly, we include rainfall and temperature shocks from one year forward as controls to ensure that our instrument remains free from contamination due to serial correlation in the weather measures. The coefficient of interest for $rainfall_{t-1}$ remains largely unchanged. Our estimates suggest that if a municipality experiences annual rainfall that is 10% below the historical average, it will see an increase of 1 percentage point in the migration outflow rate.

Next, we distribute the predicted migration outflow shock based on past settlement patterns of migrants moving from the origin municipality, o, to the destination municipality, d. A *sine qua non* requirement implicit in our empirical framework is that both the predicted migration outflow and inflow rates, denoted by $\widehat{m}_{ot}$ and $\widehat{m}_{dt}$ respectively, should be strongly correlated with their observed values. Figure 2 demonstrates that our predictions closely match the observed migration patterns. Panel (a) illustrates the relationship between the predicted and observed number of migrants leaving the Semiarid region and entering non-Semiarid municipalities from 1996 to 2010. Panel (b) presents the predicted and observed numbers of incoming migrants to Semiarid municipalities.

In Appendix G, we provide a more detailed description of our data source for weather shocks, discuss alternative weather measures, and present additional information on how we constructed our instrument, including past and predicted settlement patterns.

This analysis indicates that our strategy yields a strong first stage, as the predicted migration rates, $\widehat{m}_{dt}$, are highly correlated with the observed migration patterns. Appendix Table H1 reveals that our first-stage point estimates are close to a one-to-one relationship (0.93), making the magnitude of the reduced-form and IV estimates almost identical, with an F-statistic of 2,079.[22]

[21] Although these shocks are similar, they are not identical, as they are computed by averaging the shocks from the grid's four nodes, weighted by the inverse distance from each node to the municipality centroid. Consequently, two municipalities within the same grid can experience differing shocks due to their varying distances from the centroid.

[22] A sufficiently high F-statistic alleviates concerns about weak instruments, particularly in light of recent discussions in Lee et al. (2020), which demonstrate that a 5 percent test requires an F-statistic of 104.7, significantly exceeding the commonly accepted threshold of 10.

5 Labor Market Effects of Migration Inflows

We now turn our attention to labor markets in the destination areas and investigate how migration inflows affect the earnings and employment of native workers. Additionally, we explore how labor markets adjust to migration shocks concerning non-wage compensation.

First, we examine how the earnings of native workers adjust to exogenous migration inflows. Table 5 presents the OLS and SSIV estimates for employment (columns 1-3) and earnings (columns 4-6). In all regressions, we control for destination-level characteristics measured in 1991, including the log of the working-age native population, the shares of the population aged 15-25, 26-50, 51-65, and older than 65; the share of the non-white population; the share of the population with a college education; the share of women in the total and employed populations; the shares of employment in agriculture and manufacturing; the logs of average household income and size; and the shares of households with access to electricity and piped water. The regressions are weighted by the working-age native population in 1991, and standard errors are clustered at the origin municipality level.

Panel A shows that the OLS results for employment and earnings are not statistically significant. When comparing the OLS results to the SSIV results in Panel B, we find strong evidence of "moving to opportunity bias" in the OLS estimates for earnings and formal employment. This bias arises because migrants often move to areas with better labor market opportunities, which can confound OLS estimates as discussed in the literature (e.g. Kleemans and Magruder (2018)). As a result, OLS estimates of migration effects may be positively biased. The results in Panel B indicate that the inflow of Semiarid migrants negatively impacts the log earnings of native workers. Specifically, a one percentage point increase in the number of migrants (equivalent to 1.1 standard deviations as detailed in Table 1) reduces earnings by 0.95%. For native workers in formal jobs, as well as those in the informal sector (including self-employed workers), an increase in the migrant share by one percentage point leads to a 0.67% reduction in formal earnings and a 0.81% reduction in informal earnings. The larger negative effect on informal earnings aligns with the absence of downward wage rigidity in this sector, thereby supporting classic predictions from perfect competition.

Our estimates for the formal sector are more significant than those found by Imbert et al. (2022), who reported a reduction of 0.15% (including pensions and housing benefits). A key difference between these approaches is that their estimates derive from changes over six years, while we utilize year-on-year changes. Similarly, our estimates for the formal sector are larger than those from Kleemans and Magruder (2018), who observed no effects despite the fact that migrants in Indonesia were more educated than natives. However, they reported larger negative effects on the wages of informal workers. Their analysis focused on the log income per hour, potentially capturing adjustments in hours worked, while we concentrate on log monthly earnings.

We also investigate the differential effects based on the position of native workers in the earnings distribution. We calculate the municipality-level thresholds for each decile of the earnings distribution, separately for each sector, and run a regression on the changes of each threshold. Figure 3 illustrates our findings. For formal sector workers, the impact appears to be smaller at the bottom of the distribution, which supports the idea of wage rigidity in this sector, thereby limiting adverse effects for low-paid workers. The entry of low-skilled migrants competing for lower-paid jobs in the formal sector can reverberate and create amplified effects upward in the earnings distribution if low and high-skill workers are seen as substitutes. Conversely, for informal workers, the impact is more pronounced for those in the bottom third of the distribution, consistent with classic predictions from perfect competition and greater substitutability between migrants and less skilled natives in this sector.

Panel B results for employment indicate no overall effect on employment levels. However, we observe offsetting employment responses between the formal and informal sectors. A one percentage point increase in migrant inflow reduces the share of formal employment by 0.13 percentage points while increasing the share of informal employment by nearly the same amount (0.11 percentage points). In contrast, Kleemans and Magruder (2018) found a larger negative effect on formal employment (-0.33 percentage points) and no significant impact on the informal sector. Our findings suggest that there is a reallocation of workers between sectors, which aligns with the significant cross-effects we observe in the formal sector, as predicted by our model concerning high substitutability between formal and informal inputs.

To provide a more comprehensive analysis, we also examine the impacts on unemployment and labor force participation, as reported in Appendix Table D1. An increase of 1 percentage point in the migration share is associated with a rise of 0.10 percentage points in the unemployment rate and a decrease of 0.09 percentage points in the proportion of individuals not participating in the labor force. One possible explanation for these findings is that if the primary earner in a household loses their job due to increased competition, other members of the household may seek employment, a phenomenon known as the added worker effect (Lundberg, 1985). We confirm this mechanism by running the same regressions separately for individuals identified as head or non-head of the household. According to Appendix Table D2, almost all employment effects come from the head of households, while the changes in unemployment and inactivity rates stem from non-head members.

5.1 Non-wage Compensation.

We now examine an additional margin of adjustment in response to migration shocks. Because firms in the formal sector cannot lower wages below the legal minimum, they may respond to labor supply shocks by reducing fringe benefits, as discussed in Section 2.2. We calculate the share of working-age native workers who receive each benefit, and Table 6 presents the estimates. A one-percentage-point increase in the predicted number of migrants corresponds to a reduction in the share of workers receiving food subsidies by 0.24 percentage points, and transport and health insurance by 0.11 percentage points. This finding aligns with our model's prediction that a labor supply shock should decrease overall formal compensation, particularly when there is a high elasticity of substitution between formal and informal inputs.

Next, we enhance these estimates by investigating firms' behavior as health insurance providers for their employees. Approximately 20% of workers obtain private health insurance through their employers. In 2018, the average cost of employer-provided health insurance was R$582, which constituted 17% of total compensation for that year (ANS, 2018).[23] Rather than relying solely on survey data, we utilize firm-level administrative data on health insurance contracts obtained from the Agência Nacional de Saúde Suplementar (ANS), the Brazilian regulatory agency responsible for overseeing the private health industry. This agency provides information on every employer-sponsored contract signed since 1940. Our data includes the date the contract was signed and a unique identifier for each firm, which allows us to merge it with RAIS, an employer-employee matched dataset from the Ministry of Labor containing firm-level data on nearly all formal employment contracts.

We define an indicator variable, $y_{idt} = \mathbb{1}\,(t \geq t^s)$, for each firm i in the destination municipality d at year t, where t^s is the year when the health insurance is procured. We then estimate how migration inflow rates in destination municipality d impact changes in y_{idt} — that is, the variation in the likelihood that firm i provides health insurance to its employees. The estimated equation is as follows:

$$\Delta y_{idt} = \alpha + \beta \widehat{m}_{dt} + \phi_i + \delta_t + \mu_{idt} \tag{11}$$

where $\widehat{m}_{dt}$ is the predicted incoming migration rate; ϕ_i and δ_t are firm and year dummies. All the regressions are weighted by the number of employees in the firm in 1996, the first year in our sample.

In column 1 of Appendix Table D3, we find that firms operating in a municipality that receives more incoming migrants are, on average, less likely to provide health insurance to employees. An increase of one percentage point in the migration rate (equal to 1.1 standard deviation, as reported in Table 1) implies a 1.5p.p. decrease in the share of firms that provide health insurance, roughly the average of y_{idt}. In Columns 2-5, we restrict the sample to different bins of firm size. The effect is near zero and insignificant for firms below 100 employees, but negative and of greater magnitude for larger firms. Firms with over 100 employees are at least six times more likely to provide health insurance as compensation.

All such evidence is consistent with formal firms systematically adjusting non-wage benefits and earnings in response to labor supply shocks, reinforcing that adjustments occur in formal contracts even in the presence of informality.

[23]See Section 2.2 for more details.

5.2 Exploring the Mechanism

Our findings show that earnings decline in both sectors due to an incoming migration shock. However, as discussed in Section 3, the extent of this decline may vary depending on the level of labor market rigidity caused by minimum wages. In areas with a more binding minimum wage, the adjustment primarily occurs through changes in formal employment and informal earnings. Conversely, in regions where the minimum wage has a limited impact, the adjustment mainly affects total compensation, with minimal effects on employment. Additionally, non-wage compensation can introduce some flexibility in wages, which could potentially help mitigate job losses.

To conduct a formal test for this mechanism, we first calculate the baseline Kaitz index (the ratio of the minimum wage to the median wage) for the year 1996. We then divide our sample into two groups based on this index. Municipalities with a high minimum wage bite are defined as those above the median Kaitz index, while those with a low minimum wage bite are those below the median. Next, we apply the same SSIV specification used in Panel B of Tables 5 and 6 separately for each group. Our findings are presented in Tables 7 and 8. In regions where the minimum wage is more binding (as indicated in columns 1-3), the informal sector faces a significantly larger negative impact on earnings. Specifically, a one percentage point increase in the migration rate corresponds to a 1.8% decrease in earnings within the informal sector. On the employment front, the rise in informal sector jobs reflects the decline in formal employment, indicating that native workers who lost their formal jobs are transitioning to the informal sector. Furthermore, there is a slight decrease in the proportion of workers receiving food benefits, while other non-wage benefit categories remain unchanged. We attribute the stability in non-wage compensation to selection effects, primarily driven by a notable reduction in the share of formal employment. Our findings suggest that workers who leave formal employment are less likely to receive non-wage benefits and are more inclined to move to the informal sector, where such benefits are not available. Additionally, positive selection among workers who remain in the formal sector helps mitigate the adverse effects predicted by our model.

In contrast, when the minimum wage is less binding (as shown in columns 4-6), the impact on informal employment and earnings is minimal. Consistent with the previously discussed mechanism, adjustments in the formal sector primarily occur through changes in total compensation, including wages and non-wage benefits. A one percentage point increase in the in-migration rate results in a 0.82% decrease in earnings in the formal sector. Additionally, the share of workers receiving non-wage benefits declines by 0.26 percentage points for food subsidies, 0.14 percentage points for transport subsidies, and 0.18 percentage points for health insurance coverage.

In principle, the varying impact of how binding the minimum wage was at baseline should primarily affect low-education workers. To test this, we conducted separate regression analyses for workers with low and high education levels based on the minimum wage bite. As shown in Table 9, we find no significant impact on formal earnings for low-education workers in markets with a high minimum wage bite, suggesting that the minimum wage is primarily binding for this group. Conversely, in markets with a low minimum wage bite, formal earnings for low-education workers decrease by 1.4%, which aligns with our expectations.

For highly educated workers, the impact on earnings is negative, regardless of the binding nature of the minimum wage at baseline. Although this group should theoretically be unaffected by minimum wage laws, we observed noticeable negative earnings effects, particularly in markets with high minimum wages. This trend suggests a high level of substitutability between high- and low-education workers, a concept that will be explored further in the next section.

Our findings indicate that when firms in the formal sector cannot reduce wages, they may resort to dismissing formal workers. These displaced workers often transition to the informal sector, where average earnings are typically lower. Additionally, firms tend to replace some high-skill native workers with low-skill informal labor. A one percentage point increase in the proportion of migrants leads to a 0.20 percentage point rise in total employment among low-education workers, while employment for high-education workers declines by a similar rate.

Regarding non-wage benefits, Table 10 shows that these benefits do not adjust for low-education workers in markets where the minimum wage is highly binding. As we previously discussed, this is likely due to a selection effect driven by a decrease in the share of formal employment. Our findings suggest that this selection effect is more pronounced among low-education workers. Specifically, less educated workers who exit formal employment in markets with high minimum wage pressures are less likely to receive non-wage benefits.

As also discussed in Section 3, the adjustment of non-wage benefits is expected to be more pronounced in regions where a significant proportion of workers receive these benefits. In such areas, employers have greater flexibility to modify non-wage benefits. To investigate this phenomenon, we divided the sample into two groups based on the baseline share of workers receiving non-wage benefits: municipalities with a high share (above median) and those with a low share. Table 12 reveals that the adjustment of non-wage benefits is stronger in regions with a higher baseline percentage of recipients. This trend is particularly evident for the most costly benefits, such as food and health. In contrast, wages and employment levels tend to adjust more in regions where a lower baseline proportion of workers receives non-wage benefits, as shown in Table 11.

5.3 Sensitivity and Heterogeneity Analysis.

This section provides an overview of the robustness checks conducted to validate our key findings. Following this, we will explore the heterogeneity of our primary estimated effects, with a focus on labor market informality and the differences based on workers' education levels.

The first issue we need to address is the possibility that labor market outcomes in destinations receiving incoming migrants may have evolved differently if the migration shock had not occurred. Following the advice of Borusyak et al. (2021), we investigate the common trend assumption by regressing lagged differences in outcomes on our shift-share instrumental variable (SSIV), while using the same set of covariates as in the main specifications. The results, presented in Table E1, reveal that several of these pre-trends are correlated with the instrument. To strengthen the validity of our identification strategy, we re-estimate our main specifications, adjusting for these pre-trends. As shown in Tables E2 and E3, the results remain robust, providing additional confidence in our findings.

Other potential issue is the possibility that the shock captured by the SSIV is serially correlated, which could lead to bias due to the dynamic adjustment of past migration flows (Jaeger et al., 2018). In Tables E4 and E5, we demonstrate that nearly all the results remain robust when controlling for the predicted number of incoming migrants in $t-1$. The only exception is the effect on formal earnings, which remains negative but is not statistically significant.

Additionally, there may be concerns regarding shifts in local labor demand that could confound our results. If this were the case, migrants from areas outside the Semiarid region might be attracted to the same destinations as those from the Semiarid. In such a scenario, we would expect to see a positive correlation between the inflow of migrants from the Semiarid and migrants from other regions. To address this, we conduct a regression analysis where we regress the migration inflow rate from various regions on the predicted inflow rate of migrants from the Semiarid region. The results are presented in Table E6. Column (1) includes time and municipality fixed effects, while Column (2) adds the same set of controls utilized in our main results. The point estimates in all specifications are close to zero and not statistically significant.

Another potential threat to our identification strategy stems from the assumption that rainfall shocks in origin municipalities affect destination labor markets solely through internal migration. A violation of this assumption could occur if a negative income shock in the origin area arises from lower rainfall and reduced trade flows to certain destination regions. In such cases, industries that are more vulnerable to trade shocks, such as agriculture or manufacturing, might experience more severe impacts. To explore this issue, we present in Table E7 the coefficients from regressions of changes in log earnings based on the predicted migrant inflow rate, disaggregated by industry. Our results show no statistically significant effects on workers' earnings in the agricultural or manufacturing sectors. Instead, the observed impacts are concentrated among native workers in the service sector, who are less likely to be affected by adverse shocks in the origin municipalities.

Finally, we examine the sensitivity of our results based on the degree of aggregation of origin regions. In Appendix H, we argue that the validity of our shift-share instrument requires the origin-level shocks to be mutually uncorrelated. Given that rainfall shocks are likely correlated across smaller geographical units, we investigate this issue in Appendix I by reconstructing our instrument using larger catchment areas for the origin of migrants—specifically, microregions or mesoregions rather than municipalities[24]. First, we document that the spatial correlation among shocks decreases significantly when we consider larger areas. Second, Tables I2-I4 demonstrate that our results relating migration to rainfall, earnings, employment, and non-wage benefits remain largely unchanged. This indicates that spatial correlation among rainfall shocks in origin municipalities does not affect our findings.

Heterogeneity by informality levels. In the model described in Section 3, the parameter θ_i represents the share of exogenous informal input that can be proxied by the informality at the market level. We show that the negative effects on earnings and total compensation in both sectors increase with θ_i, and our empirical results support this prediction. Table F1 illustrates stronger effects on earnings and employment in municipalities with higher levels of baseline informality. However, in Table F2, we do not observe a larger adjustment in non-wage benefits in these municipalities. If we assume that job losses in the formal sector primarily affect workers who are less likely to receive benefits, then positive selection of workers staying in the formal sector explains why negative impacts on non-wage benefits are attenuated in labor markets characterized by high informality.

Heterogeneity by education. Next, we assess whether individuals with different levels of education experience varying impacts. In our study context, we expect low-education native workers to be close substitutes for migrants. This expectation is supported by the fact that the marginal migrant affected by the drought tends to be less skilled than the average migrant. Appendix Table G5 displays the results of a regression analyzing the change in the share of less-educated individuals in the origin municipality resulting from the drought shock, which shows evidence of negative selection. Therefore, we anticipate that Predictions (i) and (ii) in Section 3 will hold stronger for less skilled workers. We define low-education as individuals with up to 7 years of schooling. In our sample, 50% of natives fall into the low-education category. We then re-estimate the effect of migration on the local labor market outcomes for natives, separately considering those with low and high education.

Table F3 presents the employment and earnings effects segmented by education level. In columns 1-3, the employment results suggest that less-educated native individuals are more likely to transition into the informal sector employment compared to those with higher education levels. Columns 4-6 examine the differential effects on log earnings. In the formal sector, earnings among native workers decline uniformly across education levels. However, the earnings loss in the informal sector is significantly greater for low-education native workers. This finding supports the hypothesis that less-educated

[24] According to IBGE (1990), microregions are defined as "groups of economically integrated municipalities sharing borders and structures of production." Mesoregions consist of collections of microregions, not all of which share borders. The Semiarid region contains 960 municipalities, 137 microregions, and 35 mesoregions.

natives face increased competition from migrants in the informal labor market, which exacerbates their earnings losses.

Regarding adjustments in non-wage benefits, Table F4 indicates that migration inflows negatively affect nearly all types of benefits across both education groups, but effects are smaller for low-education workers. This can be attributed to positive selection; that is, low-education workers who exit the formal sector are less likely to receive benefits. Consequently, the estimates for low-education workers are potentially overestimated towards little or no adjustments in their benefits.

The impact is also negative among highly educated workers, despite the absence of direct competition from immigrants in this group. This finding aligns with existing literature in developed countries, which highlights significant cross-effects: low-skill immigration impacts the earnings of high-skill workers and vice versa, depending on the degree of substitution or complementarity between skill types (e.g., Borjas, 2003; Card and Lemieux, 2001; Beerli et al., 2021). Assuming a relatively high elasticity of substitution between low-education and high-education workers within each sector, we expect that the effects on earnings or total compensation for high-education workers will be negative, which aligns with our findings.

6 Discussion and Concluding Remarks

This paper explores the impact of weather-related internal migration on Brazil's labor market. Our methodology uses a shift-share instrument approach, which combines the variation in the number of individuals leaving their hometowns due to weather shocks with historical settlement patterns. This approach allows us to take advantage of the exogenous variation in the number of migrants arriving in each destination municipality.

Overall, our findings indicate that an influx of low-skill workers leads to a decrease in earnings within the unregulated informal sector. We also observe declines in the formal sector, with near-zero earnings estimates for low-education workers in regions with a high minimum wage bite. In these areas, reductions in formal employment are largely absorbed by informal jobs. Our estimates of non-wage benefits—such as food, transportation, and health insurance—are considered conservative due to potential selection effects, especially in areas with a high minimum wage impact and in informal markets that involve low-education workers. In these situations, the makeup of formal employment changes. If we assume positive selection (meaning that workers who are less likely to receive benefits tend to leave formal jobs), the negative impact seen on average across most markets is likely reduced. Without these selection effects, taking non-wage adjustments into account indicates that the decreases in formal compensation may be even more pronounced than analyses that do not consider these factors.

We frame our results in a context where formal and informal labor inputs are not perfect substitutes, and non-wage benefits yield predictions that closely align with our empirical findings. Our findings indicate that non-wage benefits are an important adjustment mechanism for firms, particularly in markets where fringe benefits are common, such as in many European and Latin American countries. These benefits allow employers to absorb part of the economic shocks, helping to mitigate the effects on employment levels.

References

Adão, R., M. Kolesár, and E. Morales (2019). Shift-share designs: Theory and inference. *The Quarterly Journal of Economics 134*(4), 1949–2010.

Albert, C., P. Bustos, and J. Ponticelli (2021). The effects of climate change on labor and capital reallocation. Technical report, National Bureau of Economic Research.

Almeida, R. and P. Carneiro (2012, July). Enforcement of labor regulation and informality. *American Economic Journal: Applied Economics 4*(3), 64–89.

Altonji, J. G. and D. Card (1991). The effects of immigration on the labor market outcomes of less-skilled natives. In *Immigration, trade, and the labor market*, pp. 201–234. University of Chicago Press.

ANS (2018). Painel de precificação: Planos de saúde. Agência Nacional de Saúde Suplementar, http://www.ans.gov.br/materiais-publicados/periodicos.

ANS (2019). Dados e indicadores do setor. Agência Nacional de Saúde Suplementar, http://www.ans.gov.br/perfil-do-setor/dados-e-indicadores-do-setor. Last accessed on 28/10/2020.

Arbache, J. S. (1995). Uma introdução aos salários indiretos no brasil. Anais, Encontro Nacional de Estudos do Trabalho, São Paulo. p. 1297-1322.

Arbache, J. S. and D. M. Ferreira (2001). Salários indiretos e previdência social no brasil. *Estudos Empresariais 6*(1), 37–44.

Assunção, J. and F. Chein (2016). Climate change and agricultural productivity in brazil: future perspectives. *Environment and Development Economics 21*(5), 581–602.

Azoulay, P., B. Jones, J. D. Kim, and J. Miranda (2022). Immigration and entrepreneurship in the united states. *American Economic Review: Insights 4*(1), 71–88.

Babecký, J., C. Berson, L. Fadejeva, A. Lamo, P. Marotzke, F. Martins, and P. Strzelecki (2019). Non-base wage components as a source of wage adaptability to shocks: evidence from european firms, 2010–2013. *IZA Journal of Labor Policy 8*(1), 1.

Babecký, J., P. Du Caju, T. Kosma, M. Lawless, J. Messina, and T. Rõõm (2012). How do european firms adjust their labour costs when nominal wages are rigid? *Labour Economics 19*(5), 792–801. Special Section on: Price, Wage and Employment Adjustments in 2007-2008 and Some Inferences for the Current European Crisis.

Beenstock, M., D. Felsenstein, et al. (2019). *The econometric analysis of non-stationary spatial panel data*. Springer.

Beerli, A., J. Ruffner, M. Siegenthaler, and G. Peri (2021). The abolition of immigration restrictions and the performance of firms and workers: Evidence from switzerland. *American Economic Review 111*(3), 976–1012.

Borjas, G. J. (2003). The labor demand curve is downward sloping: Reexamining the impact of immigration on the labor market. *The Quarterly Journal of Economics 118*(4), 1335–1374.

Borjas, G. J. (2014). *Immigration Economics*. Cambridge: Harvard University Press.

Borjas, G. J. and J. Monras (2017). The labour market consequences of refugee supply shocks. *Economic Policy 32*, 361—-413.

Borusyak, K., P. Hull, and X. Jaravel (2021). Quasi-experimental shift-share research designs. *The Review of Economic Studies*, Forthcoming.

Bosch, M. and J. Esteban-Pretel (2012). Job creation and job destruction in the presence of informal markets. *Journal of Development Economics 98*(2), 270–286.

Boustan, L. P., P. V. Fishback, and S. Kantor (2010). The effect of internal migration on local labor markets: American cities during the great depression. *Journal of Labor Economics 28*(4), 719–746.

Busso, M., J. P. Chauvin, and N. Herrera L (2021). Rural-urban migration at high urbanization levels. *Regional Science and Urban Economics 91*, 103658.

Carabetta, J., R. H. M. Pereira, and C. N. Goncalves (2020). geobr: Loads shapefiles of official spatial data sets of brazil. python package version 0.1.5. *The Python Package Index (PyPI)*.

Card, D. (2001). Immigrant inflows, native outflows, and the local labor market impacts of higher immigration. *Journal of Labor Economics 19*(1), 22–64.

Card, D. (2009, May). Immigration and inequality. *American Economic Review 99*(2), 1–21.

Card, D. and T. Lemieux (2001). Can falling supply explain the rising return to college for younger men? a cohort-based analysis. *The Quarterly Journal of Economics 116*(2), 705–746.

Card, D. and G. Peri (2016). Immigration Economics by George J. Borjas: A Review Essay. *Journal of Economic Literature 54*(4), 1333–49.

Cardoso, A. and T. Lage (2006). Labor inspection in Brazil. *Dados 2*, Rio de Janeiro.

Cardoso, A. and P. Portugal (2005). Contractual wages and the wage cushion under different bargaining settings. *Journal of Labor Economics 23*(4), 875–902.

Carvalho, J. A. and R. A. Garcia (2002). Estimativas decenais e quinquenais de saldos migratorios e taxas líquidas de migração do brasil. Working paper, Cedeplar/UFMG.

Clemens, J. (2021, February). How do firms respond to minimum wage increases? understanding the relevance of non-employment margins. *Journal of Economic Perspectives 35*(1), 51–72.

Clemens, J., L. B. Kahn, and J. Meer (2018, May). The minimum wage, fringe benefits, and worker welfare. Working Paper 24635, National Bureau of Economic Research.

Corbi, R., R. Ferreira, R. Narita, and D. Souza (2022). Labor courts, job search and employment: Evidence from a labor reform in brazil. Working paper, University of Sao Paulo.

Correia, S. (2017). Linear models with high-dimensional fixed effects: An efficient and feasible estimator. Working paper. http://scorreia.com/research/hdfe.pdf.

Dustmann, C., T. Frattini, and I. P. Preston (2012). The effect of immigration along the distribution of wages. *Review of Economic Studies 80*(1), 145–173.

Dustmann, C. and A. Glitz (2015). How do industries and firms respond to changes in local labor supply? *Journal of Labor Economics 33*(3), 711–50.

Dustmann, C., U. Schönberg, and J. Stuhler (2016). The impact of immigration: Why do studies reach such different results? *The Journal of Economic Perspectives 30*(4), 31–56.

El-Badaoui, E., E. Strobl, and F. Walsh (2017). Impact of internal migration on labor market outcomes of native males in thailand. *Economic Development and Cultural Change 66*(1), pp. 147–177.

Fields, G. (1975). Rural-urban migration, urban unemployment and underemployment, and job-search activity in ldcs. *Journal of Development Economics 2*(2), 165–187.

Foged, M. and G. Peri (2016). Immigrants' effect on native workers: New analysis on longitudinal data. *American Economic Journal: Applied Economics 8*(2), 1–34.

Goldsmith-Pinkham, P., I. Sorkin, and H. Swift (2020, August). Bartik instruments: What, when, why, and how. *American Economic Review 110*(8), 2586–2624.

Gonzaga, G. (2003). Labor turnover and labor legislation in Brazil. *Economía (Journal of the Latin American and Caribbean Economic Association) 4*(1), 165–222.

Harris, I., T. J. Osborn, P. Jones, and D. Lister (2020). Version 4 of the cru ts monthly high-resolution gridded multivariate climate dataset. *Scientific data 7*(1), 1–18.

Harris, J. R. and M. P. Todaro (1970). Migration, unemployment and development: A two-sector analysis. *American Economic Review 60*(1), 126–142.

Hornbeck, R. (2012). The enduring impact of the american dust bowl: Short-and long-run adjustments to environmental catastrophe. *American Economic Review 102*(4), 1477–1507.

IBGE (1990). Divisão do brasil em mesorregiões e microrregiões geográficas. Instituto Brasileiro de Geografia e Estatística, https://www.ibge.gov.br/geociencias/organizacao-do-territorio/divisao-regional/15778-divisoes-regionais-do-brasil.html?edicao=16163&t=downloads. Last accessed on 13/08/2021.

Imbert, C., M. Seror, Y. Zhang, and Y. Zylberberg (2022). Migrants and Firms: Evidence from China. *American Economic Review 112*(6), 1885–1914.

Imbert, C. and G. Ulyssea (2024). Rural Migrants and Urban Informality: Evidence from Brazil. Working paper, University College London.

Jaeger, D. A., J. Ruist, and J. Stuhler (2018, February). Shift-Share Instruments and the Impact of Immigration. Working Paper 24285, National Bureau of Economic Research.

Kahn, A. S. and R. T. Campus (1992). Effects of drought on agricultural sector of Northeast Brazil. Working paper.

Kleemans, M. and J. Magruder (2018). Labour Market Responses To Immigration: Evidence From Internal Migration Driven By Weather Shocks. *The Economic Journal 128*(613), 2032–2065.

Lagakos, D. (2020, August). Urban-rural gaps in the developing world: Does internal migration offer opportunities? *Journal of Economic Perspectives 34*(3), 174–92.

Lee, D. S., J. McCrary, M. J. Moreira, and J. Porter (2020). Valid t-ratio inference for iv. Papers, arXiv.org.

Lewis, E. (2011). Immigration, skill mix, and capital skill complementarity. *Quarterly Journal of Economics 126*(2), 1029–69.

Lundberg, S. (1985). The added worker effect. *Journal of Labor Economics 3*(1), 11–37.

Marinho, C. R. V. (2020). Essays on health economics. PhD Dissertation, Insper.

Mazumdar, D. (1976). The urban informal sector. *World Development 4*(8), 655–679.

McKenzie, R. (1980). The labor market effects of minimum wage laws: A new perspective. *Journal of Labor Research 1*, 255–264.

Meghir, C., R. Narita, and J.-M. Robin (2015, April). Wages and informality in developing countries. *American Economic Review 105*(4), 1509–46.

Molloy, R., C. L. Smith, and A. Wozniak (2011, September). Internal migration in the united states. *Journal of Economic Perspectives 25*(3), 173–96.

Munshi, K. (2003). Networks in the modern economy: Mexican migrants in the us labor market. *The Quarterly Journal of Economics 118*(2), 549–599.

Narita, R. (2020). Self-employment in developing countries: A search-equilibrium approach. *Review of Economic Dynamics 35*, 1–34.

Ottaviano, G. I. and G. Peri (2012). Rethinking the effect of immigration on wages. *Journal of the European Economic Association 10*(1), 152–197.

Ottaviano, G. I., G. Peri, and G. C. Wright (2013). Immigration, offshoring, and american jobs. *American Economic Review 103*(5), 1925–59.

Perry, G., W. Maloney, O. Arias, P. Fajnzylber, A. Mason, and J. Saavedra-Chanduvi (2007). *Informality: Exit or Exclusion*. Washington, DC: World Bank.

Rocha, R. and R. R. Soares (2015). Water scarcity and birth outcomes in the brazilian semiarid. *Journal of Development Economics 112*, 72–91.

Skoufias, E., K. Vinha, and H. Conroy (2013). The impacts of climate variability on household welfare in rural mexico. *Population and Environment 34*(3), 370–399.

Ulyssea, G. (2010). Regulation of entry, labor market institutions and the informal sector. *Journal of Development Economics 91*, 87–99.

Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from brazil. *American Economic Review 108*(8), 2015–47.

Ulyssea, G. (2020). Informality: Causes and consequences for development. *Annual Review of Economics 12*(1), 525–546.

UN DESA (2017). International migration report 2017. United Nations Department of Economic and Social Affairs, Population Division.

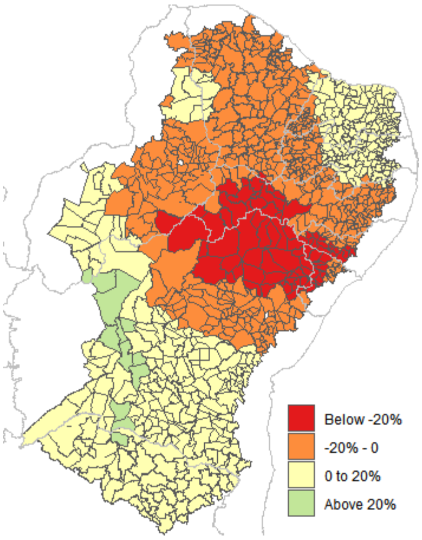
UNDP (2009). Overcoming barriers: Human mobility and development. United Nations Development Program, Human Development Report 2009. http://hdr.undp.org/en/content/human-development-report-2009. Last accessed on 03/09/2021.

Vicente-Serrano, S. M., S. Beguería, and J. I. López-Moreno (2010). A multiscalar drought index sensitive to global warming: the standardized precipitation evapotranspiration index. *Journal of Climate 23*(7), 1696–1718.

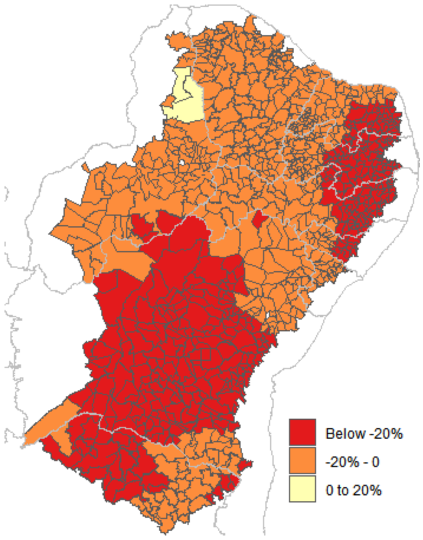
Wang, X., A. S. Auler, R. L. Edwards, H. Cheng, P. S. Cristalli, P. L. Smart, D. A. Richards, and C.-C. Shen (2004). Wet periods in northeastern brazil over the past 210 kyr linked to distant climate anomalies. *Nature 432*, 740–743.

Figures and tables

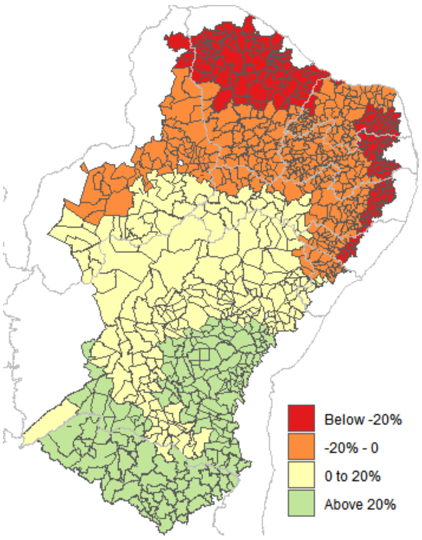
Figure 1: Precipitation levels in the Semiarid region for selected years



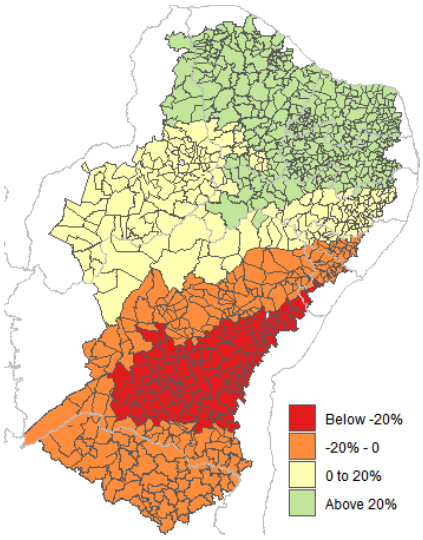
(a) 1997



(b) 2001



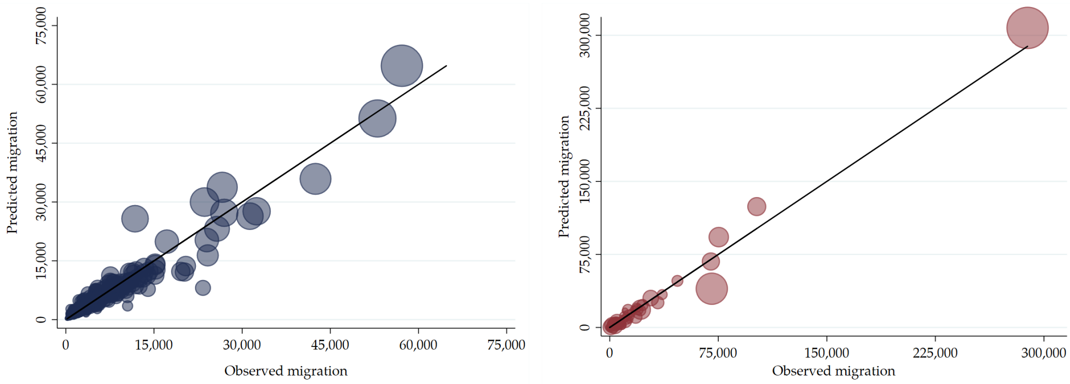
(c) 2005



(d) 2009

Notes: This figure presents rainfall distribution across the Semiarid region municipalities for selected years. Rainfall is measured as the log-deviations from historical averages. *Data source*: CRU Time Series v4 (Harris et al., 2020).

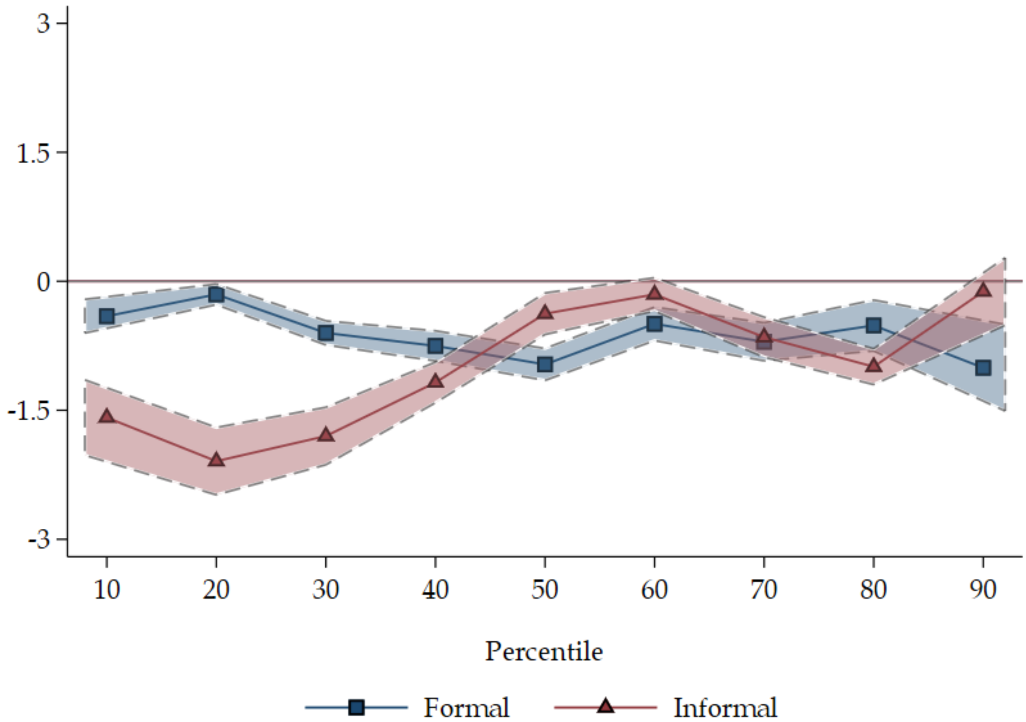
Figure 2: Observed vs predicted migration



(a) Migration outflow (b) Migration inflow

Notes: This figure presents the relationship between the accumulated predicted and observed migration flows across Brazilian municipalities from 1996 to 2010. Panel (a) shows the number of migrants leaving the Semiarid region to non-Semiarid municipalities. Panel (b) shows the number of incoming Semiarid migrants for destination municipalities. The circle size represents the municipality's total population in 1991. *Data source*: Census microdata (IBGE).

Figure 3: Effects of predicted migration along the earnings distribution



Notes: This figure plots origin-level SSIV coefficients of the incoming migration rate in the changes of the cutoffs of the earnings distribution, by sector. The informal sector also includes self-employed workers. Controls include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water).

Table 1: **Summary statistics: weather and migration data**

Panel A: Origin (Semiarid)	Mean	Std. Dev.	Min	Max	Obs
Annual Rainfall	793.22	212.84	280.85	2,061.18	14,400
Rainfall shock	-0.01	0.19	-0.72	0.48	14,400
Annual Temperature	25.55	1.38	21.76	28.93	14,400
Temperature shock	0.01	0.01	-0.01	0.05	14,400
Out-migration	214.26	323.80	0.00	5,773.00	14,400
Out-migration rate (p.p.)	1.05	0.62	0.00	7.22	14,400
Population	21,387	30,405	1,265	480,949	14,400
Panel B: Destination (Non-Semiarid)	**Mean**	**Std. Dev.**	**Min**	**Max**	**Obs**
Annual Rainfall	1,622.56	406.64	665.70	3,594.01	8,190
Rainfall shock	0.04	0.16	-0.76	0.65	8,190
Annual Temperature	23.15	2.83	15.92	28.73	8,190
Temperature shock	0.03	0.02	-0.03	0.07	8,190
In-migration	146.81	897.18	0.00	25,423.00	8,190
In-migration rate (p.p.)	0.28	0.91	0.00	24.52	8,190
Native population	56,516	249,883	320	5,135,847	8,190

Notes: Rainfall is measured in mm. Temperature is measured in degrees Celsius. Migration outflow (inflow) rate is the share of migrants over the local (native) population.

Table 2: **Summary statistics:**
Native individuals in destination municipalities

	Mean	Std. Dev.	Min	Max	Obs
Individual Characteristics					
Female	52.39	2.45	0.00	72.72	8,190
Black	7.49	5.34	0.00	53.85	8,190
White	56.94	20.53	0.00	100.00	8,190
Age	37.68	1.55	30.15	55.00	8,190
Years of schooling	7.76	1.42	0.00	13.52	8,190
Less than elementary	44.06	13.93	3.19	100.00	8,190
Employment					
Any employment	62.81	5.56	10.00	100.00	8,190
Formal	35.21	8.16	0.00	100.00	8,190
Informal	27.60	6.28	0.00	81.80	8,190
Unemployed	11.89	4.69	0.00	80.00	8,190
Out of labor force	25.30	4.97	0.00	58.14	8,190
Earnings					
Overall	849.39	362.91	60.88	3,582.08	8,190
Formal	1,017.43	417.65	58.67	15,167.10	8,174
Informal	634.58	287.89	20.00	4,941.10	8,184
Non-wage benefits					
Food	16.86	7.48	0.00	58.34	8,190
Transport	17.84	8.03	0.00	53.67	8,190
Health	9.40	5.38	0.00	100.00	8,190

Notes: Each observation is a destination municipality-year cell, weighted by its 1991 native population. Data come from PNAD in 1997-1999 and 2001-2009. Earnings are measured in R$ of 2012. The informal sector also includes self-employed workers. Non-wage benefits are percentages of the working-age native population.

Table 3: **Comparative characteristics: natives vs migrants**

	Migrants	All natives	Diff.	Low-ed. natives	High-ed. natives
Individual Characteristics					
Age	30.59	37.08	-6.49***	39.74	33.18
	(4.55)	(1.55)		(2.95)	(2.63)
Female	49.47	51.30	-1.83***	50.34	53.07
	(15.66)	(1.96)		(2.90)	(3.54)
Non-white	53.15	43.13	10.02***	49.70	35.96
	(21.06)	(22.83)		(22.87)	(22.71)
Married	59.57	62.00	-2.43***	67.17	55.44
	(18.60)	(5.13)		(5.42)	(5.11)
Number of children	2.34	2.58	-0.24***	2.97	1.79
	(1.11)	(0.53)		(0.43)	(0.40)
Share of less-educated	57.08	49.54	7.54***	100.00	0.00
	(23.75)	(17.79)		(0.00)	(0.00)
Labor market conditions					
Share of employed	65.17	61.63	3.54***	56.06	69.15
	(17.16)	(7.41)		(6.89)	(6.64)
Share of formal	36.43	32.59	3.84***	24.69	41.95
	(19.88)	(11.54)		(9.27)	(9.93)
Earnings	1016.28	1432.56	-416.28***	877.62	1834.44
	(748.12)	(628.59)		(282.09)	(740.09)
Work less than 40h/week	9.42	12.18	-2.77***	10.08	15.70
	(10.68)	(3.54)		(3.55)	(4.83)

Notes: This table shows comparisons of migrants from the Semiarid region and native individuals by education level, using data from the Census (1991, 2000, and 2010) only for the municipalities also covered by *PNAD*. Each observation is a destination municipality-year cell, weighted by its 1991 native population. We define less-educated individuals those with less than 7 years of schooling. Earnings are measured in R$ of 2012. The shares of formal employment, earnings, and worked hours are calculated only for employed individuals.

Table 4: **Migration outflows induced by weather shocks**

	(1)	(2)	(3)	(4)	(5)	(6)
Rainfall$_{t-1}$	-0.100***	-0.094***	-0.095***	-0.094***	-0.094***	-0.096***
	(0.029)	(0.028)	(0.028)	(0.028)	(0.029)	(0.030)
Rainfall$_{t-2}$			0.007	0.021		
			(0.030)	(0.031)		
Rainfall$_{t-3}$				0.062**		
				(0.028)		
Rainfall$_{t}$					-0.047	
					(0.031)	
Rainfall$_{t+1}$						-0.060*
						(0.036)
Observations	14,400	14,400	14,400	14,400	14,400	14,400
Municipalities	960	960	960	960	960	960
R-Squared	0.461	0.465	0.465	0.465	0.465	0.465
Time dummies	✓	✓	✓	✓	✓	✓
Municipality dummies	✓	✓	✓	✓	✓	✓
Temperature shocks	✓	✓	✓	✓	✓	✓
Demographics		✓	✓	✓	✓	✓

Notes: Each observation is an origin municipality-year cell. The dependent variable is the number of individuals who left the origin municipality divided by the total population in the 1991 Census. Rainfall is measured as the log-deviation from the historical average (for the 6 months in the crop growing season). All specifications include controls for temperature shocks, municipality, and year fixed effects. Columns (2)-(6) also control for municipality-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Standard errors are clustered at the grid level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 5: **Effects of migration on employment and earnings**

	Employment			Earnings		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS						
In-migration rate	-0.090	-0.029	-0.061	0.165	0.310	-0.228
	(0.091)	(0.097)	(0.082)	(0.427)	(0.466)	(0.450)
Observations	8,190	8,190	8,190	8,190	8,162	8,179
Municipalities	684	684	684	684	684	684
Panel B: SSIV						
In-migration rate	-0.014	-0.127***	0.113***	-0.954***	-0.672***	-0.808***
	(0.036)	(0.040)	(0.036)	(0.209)	(0.212)	(0.129)
Observations	11,460	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955	955

Notes: This table shows destination-level OLS estimates and origin-level SSIV coefficients by sector. Each observation is a municipality-year cell. The informal sector also includes self-employed workers. In Columns (1)-(3), the dependent variable is the change in the employment rate, while in Columns (4)-(6), it is the change in log earnings. All regressions control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) and are weighted by the working-age native population in 1991. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 6: **Effects of migration on non-wage benefits**

	Food	Transport	Health
	(1)	(2)	(3)
Panel A: OLS			
In-migration rate	-0.061	0.027	-0.048
	(0.094)	(0.083)	(0.081)
Observations	8,190	8,190	8,190
Municipalities	684	684	684
Panel B: SSIV			
In-migration rate	-0.236***	-0.113***	-0.111***
	(0.037)	(0.026)	(0.026)
Observations	11,460	11,460	11,460
Municipalities	955	955	955

Notes: This table shows destination-level OLS estimates and origin-level SSIV coefficients of changes in the proportion of working-age native workers receiving non-wage benefits. Each observation is a municipality-year cell. All regressions control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) and are weighted by the working-age native population in 1991. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 7: **Effects of migration on employment and earnings, by level of MW bite**

	High MW Bite			Low MW Bite		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Δ log earnings						
In-migration rate	-1.403***	-0.705	-1.766***	-0.738***	-0.818***	-0.209
	(0.465)	(0.524)	(0.241)	(0.185)	(0.168)	(0.141)
Panel B: Δ employment rate						
In-migration rate	0.005	-0.323***	0.328***	-0.020	-0.079***	0.059
	(0.067)	(0.104)	(0.065)	(0.038)	(0.026)	(0.049)
Observations	9,840	9,840	9,840	11,460	11,460	11,460
Municipalities	820	820	820	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in employment rate and log earnings for each sector and by level of minimum wage bite. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All regressions include time dummies, control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water), and are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 8: **Effects of migration on non-wage benefits, by level of MW bite**

	High MW Bite			Low MW Bite		
	Food	Transport	Health	Food	Transport	Health
	(1)	(2)	(3)	(4)	(5)	(6)
In-migration rate	-0.108*	-0.015	0.039	-0.257***	-0.138***	-0.179***
	(0.058)	(0.043)	(0.030)	(0.035)	(0.030)	(0.025)
Observations	9,840	9,840	9,840	11,460	11,460	11,460
Municipalities	820	820	820	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in the proportions of working-age native population who receive health insurance, food, or transport subsidies by level of minimum wage bite. Each observation is an origin municipality-year cell. All regressions include time dummies, control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water), and are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 9: **Effects of migration on employment and earnings, by level of MW bite and education**

	High MW Bite			Low MW Bite		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Δ log earnings (low-education)						
In-migration rate	-0.889***	-0.071	-1.273***	-1.427***	-1.435***	-0.946***
	(0.176)	(0.256)	(0.205)	(0.138)	(0.159)	(0.181)
Panel B: Δ log earnings (high-education)						
In-migration rate	-2.430***	-1.340**	-3.462***	-0.426**	-0.676***	0.276
	(0.642)	(0.622)	(0.491)	(0.197)	(0.189)	(0.180)
Panel C: Δ employment rate (low-education)						
In-migration rate	0.201***	-0.100***	0.301***	0.014	-0.080***	0.094***
	(0.057)	(0.035)	(0.069)	(0.040)	(0.020)	(0.032)
Panel D: Δ employment rate (high-education)						
In-migration rate	-0.196**	-0.223***	0.027	-0.033	0.001	-0.035
	(0.091)	(0.079)	(0.022)	(0.031)	(0.029)	(0.022)
Observations	9,840	9,840	9,840	11,460	11,460	11,460
Municipalities	820	820	820	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in employment rate and log earnings for each sector and by the minimum wage bite and education levels. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All regressions include time dummies, control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access

Table 10: **Effects of migration on non-wage benefits, by level of MW bite**

	High MW Bite			Low MW Bite		
	Food	Transport	Health	Food	Transport	Health
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Δ benefits (low-education)						
In-migration rate	-0.036	0.027	0.048***	-0.121***	-0.065***	-0.035***
	(0.031)	(0.025)	(0.015)	(0.009)	(0.013)	(0.008)
Panel B: Δ benefits (high-education)						
In-migration rate	-0.071**	-0.042	-0.009	-0.136***	-0.073***	-0.143***
	(0.030)	(0.032)	(0.017)	(0.034)	(0.024)	(0.020)
Observations	9,840	9,840	9,840	11,460	11,460	11,460
Municipalities	820	820	820	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in the proportions of working-age native population who receive health insurance, food, or transport subsidies by the minimum wage bite and education levels. Each observation is an origin municipality-year cell. All regressions include time dummies, control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water), and are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 11: **Effects of migration on employment and earnings, by baseline level of non-wage benefits**

	High share of non-wage benef.			Low share of non-wage benef.		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Δ log earnings						
In-migration rate	-0.452**	-0.449*	-0.176	-2.505***	-1.694***	-2.372***
	(0.229)	(0.239)	(0.142)	(0.294)	(0.397)	(0.253)
Panel B: Δ employment rate						
In-migration rate	-0.063	-0.107**	0.045	-0.072	-0.401***	0.329***
	(0.043)	(0.049)	(0.042)	(0.056)	(0.063)	(0.060)
Observations	11,460	11,460	11,460	9,192	9,192	9,192
Municipalities	955	955	955	766	766	766

Notes: This table shows origin-level SSIV coefficients on changes in employment rate and log earnings for each sector and by baseline level of non-wage benefits. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All regressions are weighted by the working-age native population in 1991, include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 12: **Effects of migration on non-wage benefits, by baseline level of non-wage benefits**

	High share of non-wage benef.			Low share of non-wage benef.		
	Food	Transport	Health	Food	Transport	Health
	(1)	(2)	(3)	(4)	(5)	(6)
In-migration rate	-0.298***	-0.107***	-0.157***	-0.196***	-0.364***	-0.142***
	(0.045)	(0.029)	(0.034)	(0.038)	(0.052)	(0.026)
Observations	11,460	11,460	11,460	9,192	9,192	9,192
Municipalities	955	955	955	766	766	766

Notes: This table shows origin-level SSIV coefficients on changes in the proportion of the working-age native population receiving non-wage benefits, by baseline level of non-wage benefits. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All regressions are weighted by the working-age native population in 1991, include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Internal Migration and Labor Market Adjustments in the Presence of Non-wage Compensation

Raphael Corbi, Tiago Ferraz, and Renata Narita

ONLINE APPENDIX

Appendix A Proofs

In this section, we provide the proofs of predictions for the effects of migration inflows from the Semiarid regions on the informal and formal sectors at the destination, as discussed in Section 3.

A.1 Prediction (i)

$$\frac{\partial W_i}{\partial L_i} < 0.$$

By differentiating (4) with respect to L_i, we obtain:

$$\frac{\partial W_i}{\partial L_i} = \alpha\theta_i K^{1-\alpha}(\theta_f L_f^\nu + \theta_i L_i^\nu)^{\frac{\alpha-2\nu}{\nu}} L_i^{\nu-2}\left[(\alpha-\nu)\theta_i L_i^\nu - (1-\nu)(\theta_f L_f^\nu + \theta_i L_i^\nu)\right] < 0$$

$$(\alpha-\nu)\theta_i L_i^\nu < (1-\nu)(\theta_f L_f^\nu + \theta_i L_i^\nu)$$

which is always true since $0 < \alpha < 1$, and $\theta_i L_i^\nu < \theta_f L_f^\nu + \theta_i L_i^\nu$.

A.2 Prediction (ii)

$$\alpha - \nu < 0 \Rightarrow \frac{\partial W_f}{\partial L_i} < 0.$$

Now, differentiating (3) with respect to L_i,

$$\frac{\partial W_f}{\partial L_i} = \frac{\alpha(\alpha-\nu)\theta_f\theta_i K^{1-\alpha}(\theta_f L_f^\nu + \theta_i L_i^\nu)^{\frac{\alpha-2\nu}{\nu}}}{(L_f L_i)^{1-\nu}}$$

which is necessarily negative if $\alpha - \nu < 0$. In the presence of binding minimum wages, $\frac{\partial B}{\partial L_i} < 0$.

Moreover, the adverse effects on earnings or total compensation are increasing in the informality share, θ_i, as the elasticities show:

$$\frac{\partial W_i}{\partial L_i}\frac{L_i}{W_i} = \frac{(\alpha-\nu)\theta_i L_i^\nu}{(\theta_f L_f^\nu + \theta_i L_i^\nu)} - (1-\nu)$$

$$\frac{\partial W_f}{\partial L_i}\frac{L_i}{W_f} = \frac{(\alpha-\nu)\theta_i L_i^\nu}{(\theta_f L_f^\nu + \theta_i L_i^\nu)}$$

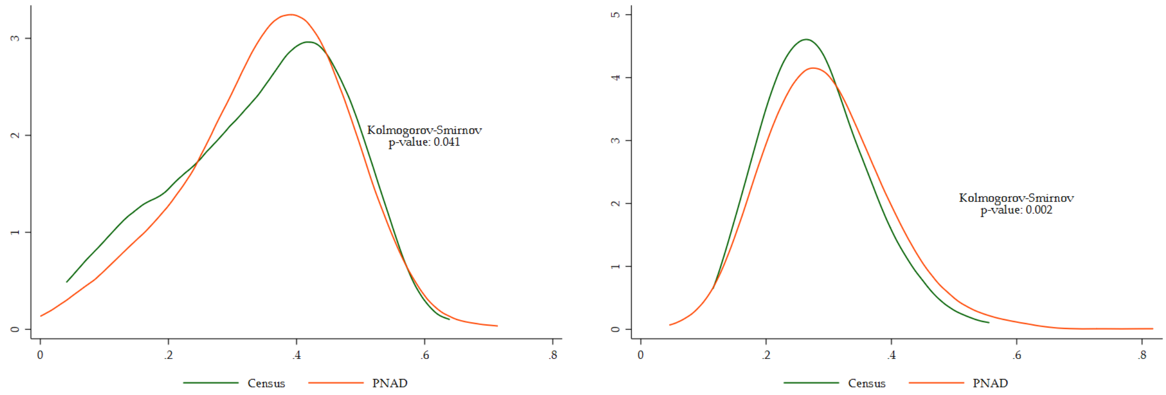
Appendix B Comparing PNAD vs Census data

Table B1: **Summary statistics: Native individuals in destination municipalities (PNAD vs Census)**

	Census	PNAD
Individual Characteristics		
Age	37.06	37.67
	(1.48)	(1.63)
Female	0.52	0.52
	(0.01)	(0.03)
Non-white	42.89	42.28
	(20.55)	(21.06)
Married	60.64	75.40
	(4.63)	(5.79)
Number of children	2.41	1.03
	(0.41)	(0.20)
Less than elementary	44.26	45.97
	(15.61)	(14.87)
Labor market conditions		
Share of employed	62.51	62.65
	(5.91)	(5.51)
Share of formal	36.11	35.62
	(8.78)	(8.38)
Earnings	1654.58	837.10
	(596.95)	(402.97)
Earnings (formal)	1729.63	985.78
	(636.75)	(456.04)
Earnings (informal)	1383.17	641.27
	(542.40)	(329.27)
Work less than 40h/week	12.60	15.90
	(3.23)	(5.81)

Notes: Each observation is a destination municipality-year cell, weighted by its 1991 native population. PNAD data from 1996, 2001, and 2009 and Census from 1991, 2000, and 2010, only for the municipalities also covered by *PNAD*. Standard deviations are in parentheses.

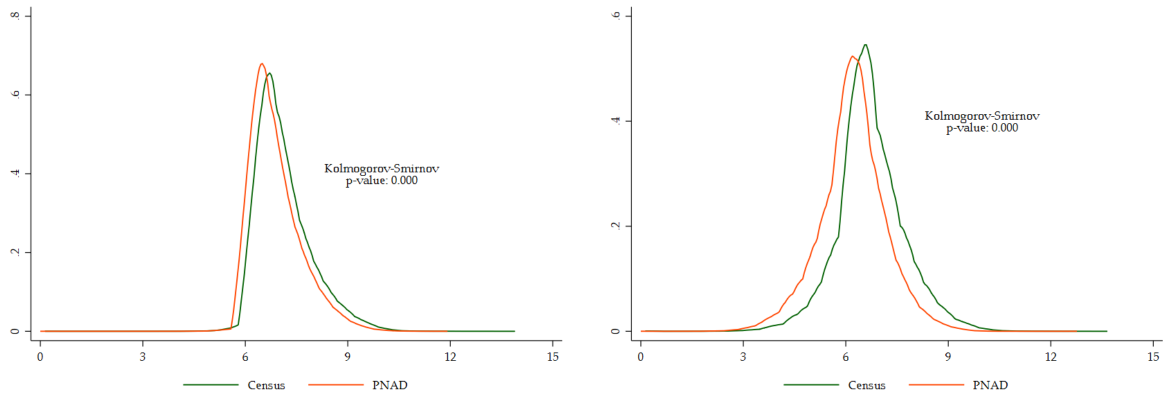
Figure B1: Distribution of employment rates, PNAD vs. Census



(a) Formal sector (b) Informal sector

Notes: Municipality-level employment rate in the formal and informal sector from PNAD (2009) and Census (2010).

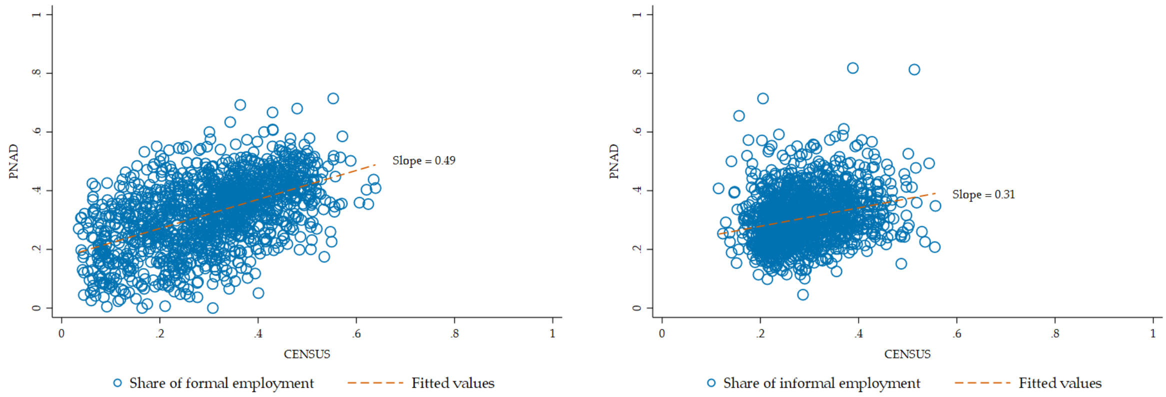
Figure B2: Distribution of log earnings, PNAD vs. Census



(a) Formal sector (b) Informal sector

Notes: Individual-level log earnings in the formal and informal sector from PNAD (2009) and Census (2010).

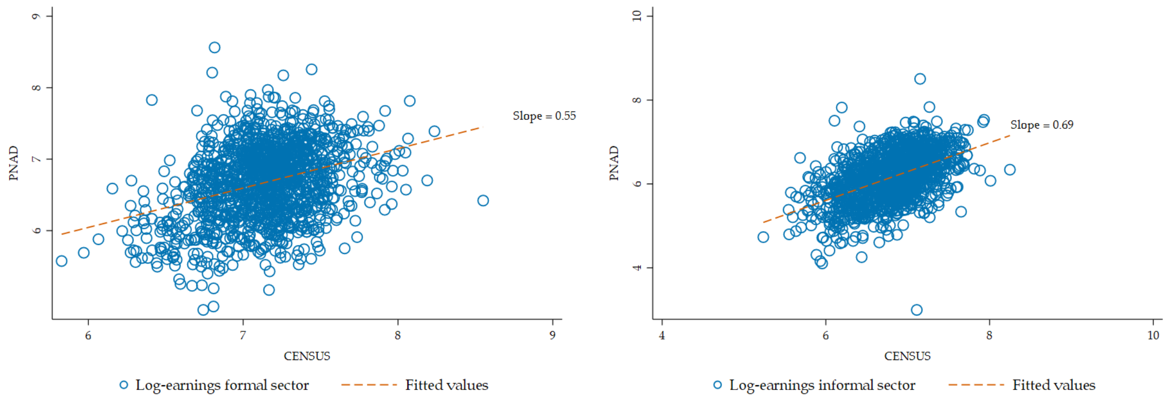
Figure B3: Relationship between PNAD and Census employment rates



(a) Formal sector (b) Informal sector

Notes: Relationship between municipality-level employment rate in the formal and informal sector from PNAD (1996, 2001, and 2009) and Census (1991, 2000, and 2010).

Figure B4: Relationship between PNAD and Census log earnings



(a) Formal sector (b) Informal sector

Notes: Relationship between municipality-level log earnings in the formal and informal sector from PNAD (1996, 2001, and 2009) and Census (1991, 2000, and 2010).

Appendix C Migrant flows from the Semiarid region

In this section, we discuss our measure of migration between cities in more detail and how we structure a yearly panel dataset from the 2000 and 2010 censuses.

C.1 Migration from the Semiarid region

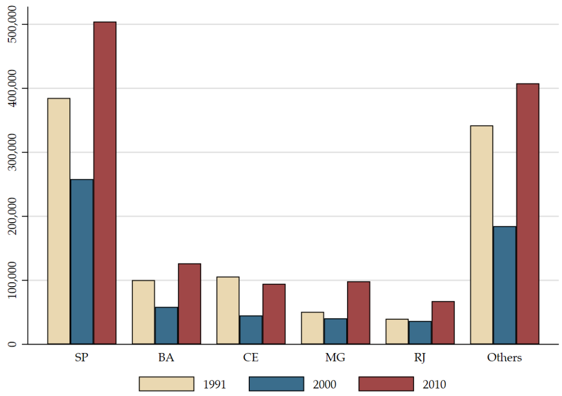
In every round of the Census, two questions allow us to track migrants by establishing their municipalities of origin and destination, as well as the year they moved. First, in the 2010 Census, respondents were asked how many years they had lived in their current municipality (ranging from one to ten years). This information enables us to determine the year of migration. We define a migrant as someone who moved to their current municipality within the previous ten years. In the 2000 Census, participants were asked about the municipality where they lived five years ago rather than their most recent place of residence. As a result, we can only identify migrants who moved as far back as 1996. This limitation is not a major concern for our analysis since 1996 is the first year for which data from the National Household Sample Survey (PNAD)—the source we use for labor market outcomes—is available. Second, respondents were also asked about the municipality they lived in before their current residence. Thus, if an individual migrated from a municipality in the Semiarid region, they will be classified as a Semiarid migrant. However, a limitation is that we can only track one origin location for each person, typically the last municipality where they lived.

The Semiarid region has historically been a significant source of migrants to other parts of the country. As shown in Figure C1, these migrants tend to be concentrated in certain states. Over the past four decades, São Paulo has alone received more than 30 percent of the migrants from the Semiarid region. In relative terms, the influx of these migrants has led to a population increase of over 2

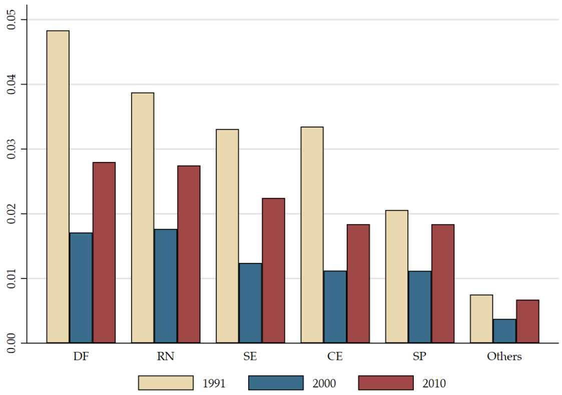
Table 3 compares migrants with both low-education and high-education natives. Migrants earn slightly less than low-education natives and have a similar likelihood of working part-time. However, they are more likely to be employed in the formal sector compared to low-education natives. In contrast, high-education natives are more likely to work in the formal sector and earn considerably higher wages than migrants from the Semiarid region.

Table C1 reveals that the top occupations for migrants — such as bricklayer for men and domestic worker for women — are also common among low-education natives, but not among skilled workers. Additionally, the same five industries that account for over 80% of working migrants also employ a similar proportion of low-education workers (see Table C2).

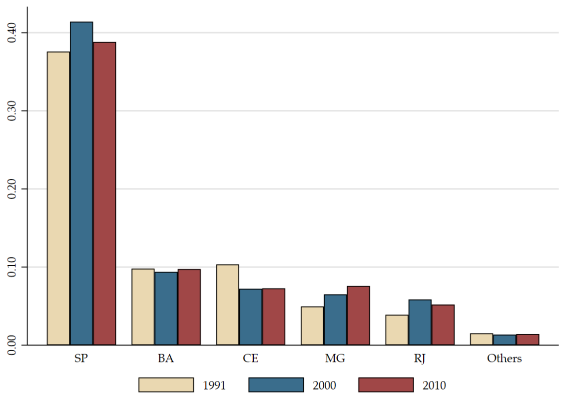
Figure C1: Top destinations for migrants from the Semiarid region



(a) Absolute number of Semiarid's migrants



(b) Semiarid's migrants as a fraction of total population



(c) Semiarid's migrants as a share of total migration

Notes: This figure illustrates the primary destination states selected by migrants from the Semiarid region. Panel (a) displays the absolute number of migrants leaving the Semiarid region for non-Semiarid areas. Panel (b) shows the same inflow as a percentage of the total population in each state, while Panel (c) presents this number as a share of the overall migrant population in each state. In each panel, the states are ranked based on their respective averages over the years. *Data source*: Census microdata (IBGE).

Table C1: **Main occupations for employed people: Migrants vs Natives**

	Position	Occupation	Share of employment	Cumulative
Migrants	1	Domestic worker	12.5	12.5
	2	Bricklayer	11.0	23.5
	3	Retail worker	10.4	33.9
	4	Rural worker	8.0	41.9
	5	Low-level office	4.2	46.1
	6	Military	4.0	50.1
	7	Non-specified occupations	2.5	52.6
	8	Manager	2.3	54.9
	9	Tailor	2.2	57.1
	10	Janitor	2.1	59.2
Low-ed. Natives	1	Rural worker	22.3	22.3
	2	Bricklayer	10.6	33.0
	3	Domestic worker	10.1	43.1
	4	Retail worker	7.8	50.9
	5	Tailor	2.9	53.7
	6	Military	2.7	56.4
	7	Manager	1.9	58.3
	8	Driver	1.8	60.1
	9	Janitor	1.7	61.8
	10	Low-level office	1.7	63.5
High-ed. Natives	1	Retail worker	10.5	10.5
	2	Low-level office	10.3	20.8
	3	Manager	7.2	28.0
	4	Teacher	6.2	34.2
	5	Military	5.6	39.8
	6	Bricklayer	3.2	43.0
	7	Rural worker	3.2	46.2
	8	Domestic worker	3.0	49.2
	9	Tailor	1.8	51.0
	10	Non-specified occupations	0.7	51.7

Notes: This table presents the top ten occupations for workers in the destination municipalities, using data from the Census (1991, 2000, and 2010).

Table C2: **Main industries for employed people: Migrants vs Natives**

	Position	Occupation	Share of employment	Cumulative
Migrants	1	Other Services	20.2	20.2
	2	Manufacturing	14.4	34.6
	3	Retail	12.8	47.4
	4	Transport/Communication	12.8	60.2
	5	Agriculture/Mining	8.9	69.1
	6	Hospitality	6.4	75.5
	7	Construction	6.3	81.8
	8	Health/Education	3.1	84.9
	9	Public Sector	0.9	85.8
Low-ed. Natives	1	Agriculture/Mining	24.0	24.0
	2	Other Services	16.0	40.0
	3	Manufacturing	13.0	53.0
	4	Retail	11.1	64.1
	5	Transport/Communication	11.1	75.1
	6	Construction	4.8	79.9
	7	Hospitality	4.2	84.1
	8	Health/Education	2.1	86.2
	9	Public Sector	1.0	87.2
High-ed. Natives	1	Other Services	32.0	32.0
	2	Transport/Communication	17.4	49.4
	3	Manufacturing	12.4	61.8
	4	Health/Education	8.3	70.1
	5	Retail	6.5	76.6
	6	Agriculture/Mining	4.0	80.6
	7	Hospitality	3.6	84.2
	8	Construction	2.9	87.0
	9	Public Sector	1.6	88.6

Notes: This table presents the top ten industries for workers in the destination municipalities, using data from the Census (1991, 2000, and 2010).

Appendix D Additional results

Table D1: **Effects of migration on unemployment and participation**

	OLS		SSIV	
	Unemployed	Inactive	Unemployed	Inactive
	(1)	(2)	(3)	(4)
In-migration rate	0.122	-0.032	0.100***	-0.086***
	(0.083)	(0.073)	(0.021)	(0.031)
Observations	8,190	8,190	11,460	11,460
Municipalities	684	684	955	955

Notes: This table shows destination-level OLS and origin-level SSIV coefficients on changes in unemployment and inactivity rates. Each observation is an origin municipality-year cell. Columns (1)-(2) show OLS estimates, while Columns (3)-(4) present the SSIV coefficients. All regressions are weighted by the working-age native population in 1991, include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Standard errors clustered at the respective municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table D2: **Effects of migration on labor market outcomes, by status in the household**

	Employment	Formal	Informal	Unemployment	Inactivity
	(1)	(2)	(3)	(4)	(5)
Panel A: Head					
In-migration rate	-0.027*	-0.118***	0.091***	0.018	0.034**
	(0.016)	(0.023)	(0.020)	(0.012)	(0.014)
Observations	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955
Panel B: Non-Head					
In-migration rate	0.013	-0.008	0.022	0.082***	-0.120***
	(0.025)	(0.020)	(0.020)	(0.015)	(0.020)
Observations	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in employment (by sector), unemployment, and inactivity rates. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. In Panel A, we use only individuals identified as the head of the household; in Panel B, only those identified as non-head are used. All regressions are weighted by the working-age native population in 1991, include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table D3: Effects of predicted in-migration on employer-provided health insurance

	(1)	(2)	(3)	(4)	(5)
Predicted inflow	-0.015**	0.003	-0.004	-0.010**	-0.048**
	(0.007)	(0.002)	(0.003)	(0.005)	(0.022)
Mean of dep. var.	0.0158	0.0131	0.0448	0.0609	0.0758
Observations	4462024	4167576	138544	142072	13832
Municipalities	682	679	480	607	280
Firms	318716	297684	9896	10148	988
Time dummies	✓	✓	✓	✓	✓
Firm dummies	✓	✓	✓	✓	✓
Firm size weighted	All firms	1 to 50	51 to 100	101 to 1000	More than 1000

Notes: This table shows the reduced form coefficients of changes in the probability of a firm offering health insurance to its employees on the predicted inflow of migrants from the Semiarid region. Each observation is a firm-year cell. The dependent variable is the difference in the dummy variable that is equal to one for every year greater than or equal to the year when the health insurance contract was signed. The regressor is the predicted number of migrants from the Semiarid region in each destination municipality (excluding those in the Semiarid region), measured as a fraction of the native working-age population in 1991. Our sample comprises a balanced panel of all firms included in RAIS during the period. All the regressions are weighted by the number of employees in the firm in 1996. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Appendix E Robustness

Table E1: **Shock balance test**

	(1)	(2)	(3)
Panel A: Lagged Δ log earnings	Overall	Formal	Informal
In-migration rate	-1.209***	-0.550***	-1.406***
	(0.199)	(0.211)	(0.132)
Panel B: Lagged Δ employment rate	Overall	Formal	Informal
In-migration rate	0.153***	-0.174***	0.327***
	(0.027)	(0.037)	(0.036)
Panel C: Lagged Δ benefits	Food	Transport	Health
In-migration rate	-0.133***	-0.167***	-0.012
	(0.033)	(0.028)	(0.018)
Observations	10,505	10,505	10,505
Municipalities	955	955	955

Notes: This table shows origin-level SSIV coefficients of the lagged change in outcomes against the predicted in-migration rate. Each observation is a municipality-year cell. The informal sector also includes self-employed workers. All regressions control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) and are weighted by the working-age native population in 1991. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table E2: **Effects of migration on employment and earnings - controlling for pre-trends**

	Employment			Earnings		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: SSIV - Without pre-trends						
In-migration rate	-0.014	-0.127***	0.113***	-0.954***	-0.672***	-0.808***
	(0.036)	(0.040)	(0.036)	(0.209)	(0.212)	(0.129)
Observations	11,460	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955	955
Panel B: SSIV - Controlling for pre-trends						
In-migration rate	-0.038	-0.148***	0.119**	-1.269***	-0.701***	-1.766***
	(0.040)	(0.044)	(0.051)	(0.283)	(0.267)	(0.216)
Observations	10,505	10,505	10,505	10,505	10,505	10,505
Municipalities	955	955	955	955	955	955

Notes: This table shows origin-level SSIV coefficients by sector. Each observation is a municipality-year cell. Panel A shows results from our baseline specification. Panel B uses annual data from 1998-1999 and 2001-2009, i.e., without the first year of our main sample, as we add the lagged outcome as an additional control. This specification uses annual data from 1998-1999 and 2001-2009. In Columns (1)-(3), the dependent variable is the change in the employment rate, while in Columns (4)-(6), it is the change in log earnings. The informal sector also includes self-employed workers. All regressions control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) and are weighted by the working-age native population in 1991. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table E3: **Effects of migration on non-wage benefits - controlling for pre-trends**

	Food	Transport	Health
	(1)	(2)	(3)
Panel A: SSIV - Without pre-trends			
In-migration rate	-0.236***	-0.113***	-0.111***
	(0.037)	(0.026)	(0.026)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Panel B: SSIV - Controlling for pre-trends			
In-migration rate	-0.400***	-0.196***	-0.127***
	(0.043)	(0.034)	(0.028)
Observations	10,505	10,505	10,505
Municipalities	955	955	955

Notes: This table shows origin-level SSIV coefficients of changes in the proportion of working-age native workers receiving non-wage benefits. Each observation is a municipality-year cell. Panel A shows results from our baseline specification. Panel B uses annual data from 1998-1999 and 2001-2009, i.e., without the first year of our main sample, as we add the lagged outcome as an additional control. In Columns (1)-(3), the dependent variable is the change in the employment rate, while in Columns (4)-(6), it is the change in log earnings. The informal sector also includes self-employed workers. All regressions control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) and are weighted by the working-age native population in 1991. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table E4: **Effects of migration on employment and earnings - controlling for serial correlation**

	Employment			Earnings		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: SSIV - Main specification						
In-migration rate	-0.014	-0.127***	0.113***	-0.954***	-0.672***	-0.808***
	(0.036)	(0.040)	(0.036)	(0.209)	(0.212)	(0.129)
Observations	11,460	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955	955
Panel B: SSIV - Controlling for lagged migration						
In-migration rate	0.012	-0.118***	0.130***	-0.617***	-0.122	-1.006***
	(0.041)	(0.044)	(0.040)	(0.223)	(0.223)	(0.147)
Observations	11,460	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955	955

Notes: This table shows origin-level SSIV coefficients by sector. Each observation is a municipality-year cell. Panel A shows results from our baseline specification. In Panel B, we add the lagged predicted migration as an additional control. In Columns (1)-(3), the dependent variable is the change in the employment rate, while in Columns (4)-(6), it is the change in log earnings. The informal sector also includes self-employed workers. All regressions control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) and are weighted by the working-age native population in 1991. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table E5: **Effects of migration on non-wage benefits - controlling for serial correlation**

	Food	Transport	Health
	(1)	(2)	(3)
Panel A: SSIV - Main specification			
In-migration rate	-0.236***	-0.113***	-0.111***
	(0.037)	(0.026)	(0.026)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Panel B: SSIV - Controlling for lagged migration			
In-migration rate	-0.270***	-0.278***	-0.151***
	(0.041)	(0.027)	(0.028)
Observations	11,460	11,460	11,460
Municipalities	955	955	955

Notes: This table shows origin-level SSIV coefficients of changes in the proportion of working-age native workers receiving non-wage benefits. Each observation is a municipality-year cell. Panel A shows results from our baseline specification. In Panel B, we add the lagged predicted migration as an additional control. All regressions control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) and are weighted by the working-age native population in 1991. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table E6: **Correlation between predicted migration from the Semiarid and other regions**

	(1)	(2)
	Non-Semiarid migration	
Predicted inflow	0.167	0.149
	(2.855)	(2.858)
Observations	8,190	8,190
Municipalities	684	684
Time dummies	✓	✓
Municipality dummies	✓	✓
Baseline $\times$ time		✓

Notes: This table shows destination-level regression coefficients of the observed inflow of migrants from other regions on the predicted number of migrants from the Semiarid, both measured as a fraction of the working-age native population in 1991. Each observation is a destination municipality-year cell. All regressions are weighted by the working-age native population in 1991 and include municipality and time dummies. Column (2) controls for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Standard errors clustered at the destination municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table E7: **Effects of migration on earnings, by industry**

	Agriculture	Manufacturing	Services
	(1)	(2)	(3)
In-migration rate	-0.455	-0.264	-0.920***
	(0.402)	(0.200)	(0.199)
Observations	11,447	11,460	11,460
Municipalities	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in log earnings by industry. Each observation is an origin municipality-year cell. All regressions are weighted by the working-age native population in 1991, include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Appendix F Heterogeneity

Table F1: **Effects of migration on employment and earnings, by baseline level of informality**

	High informality			Low informality		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Δ log earnings						
In-migration rate	-2.996***	-1.811***	-3.454***	-0.683**	-0.808***	-0.180
	(0.245)	(0.350)	(0.225)	(0.266)	(0.277)	(0.162)
Panel B: Δ employment rate						
In-migration rate	-0.002	-0.291***	0.289***	-0.063	-0.133**	0.070
	(0.049)	(0.057)	(0.048)	(0.050)	(0.055)	(0.045)
Observations	9,696	9,696	9,696	11,460	11,460	11,460
Municipalities	808	808	808	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in employment rate and log earnings for each sector and by baseline level of informality. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All regressions include time dummies, control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water), and are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table F2: **Effects of migration on non-wage benefits, by baseline level of informality**

	High informality			Low informality		
	Food	Transport	Health	Food	Transport	Health
	(1)	(2)	(3)	(4)	(5)	(6)
In-migration rate	-0.261***	-0.404***	-0.076***	-0.282***	-0.071*	-0.127***
	(0.036)	(0.052)	(0.029)	(0.055)	(0.036)	(0.040)
Observations	9,696	9,696	9,696	11,460	11,460	11,460
Municipalities	808	808	808	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in the proportions of working-age native population who receive health insurance, food, or transport subsidies by baseline level of informality. Each observation is an origin municipality-year cell. All regressions include time dummies, control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water), and are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table F3: **Effects of migration on employment and earnings, by education level**

	Employment			Earnings		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Low-Education						
In-migration rate	0.087***	-0.059***	0.146***	-1.363***	-0.988***	-1.260***
	(0.031)	(0.016)	(0.031)	(0.115)	(0.136)	(0.123)
Observations	11,460	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955	955
Panel B: High-education						
In-migration rate	-0.101**	-0.067*	-0.033**	-0.938***	-0.798***	-0.730***
	(0.042)	(0.036)	(0.015)	(0.250)	(0.245)	(0.189)
Observations	11,460	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in employment rate and log earnings for each sector and by education level. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All regressions are weighted by the working-age native population in 1991, include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table F4: **Effects of migration on non-wage benefits, by education level**

	Food	Transport	Health
	(1)	(2)	(3)
Panel A: Low-Education			
In-migration rate	-0.098***	-0.045***	0.004
	(0.010)	(0.011)	(0.007)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Panel B: High-education			
In-migration rate	-0.138***	-0.068***	-0.115***
	(0.030)	(0.020)	(0.021)
Observations	11,460	11,460	11,460
Municipalities	955	955	955

Notes: This table shows origin-level SSIV coefficients on changes in the proportion of the working-age native population receiving non-wage benefits, by education level. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All regressions are weighted by the working-age native population in 1991, include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Appendix G Weather shocks and predicted migration

This section reviews the weather data and explains how we construct our instrument. Additionally, we demonstrate that our results are consistent with an alternative measure of weather shocks.

G.1 Weather data

Our primary source of weather data is the CRUTS v4 dataset, which is produced by the Climatic Research Unit at the University of East Anglia (Harris et al., 2020). This gridded dataset provides monthly precipitation and temperature information covering the globe (excluding Antarctica) from 1901 to 2018. The grid has a resolution of 0.5° × 0.5° (approximately 56km^2) and is created through interpolation using data from ground-based weather stations worldwide.

To obtain the shapefile for Brazilian municipalities, we utilize the R package 'geobr' (Carabetta et al., 2020). We then georeference the coordinates of each municipality's centroid. For each municipality, we identify the four grid points closest to its centroid and calculate the average precipitation and temperature from these points, weighted by their inverse distance to the centroid.

This procedure generates a dataset containing monthly averages of precipitation and temperature for each municipality from 1901 to 2010, which we then aggregate into yearly measures. Precipitation is calculated as the sum of monthly levels, while temperature is determined by averaging the monthly values. For each municipality in the Semiarid region, we compute the historical mean for both variables and then take the natural logarithm of these values (including both levels and long-term averages).

Lastly, we define our weather shock variables as follows:

$$Rainfall_{ot} = \ln\left(\sum_{\tau \in \{GS\}} r_{o\tau t}\right) - \ln(\bar{r}_o) \tag{G1}$$

where $r_{o\tau t}$ represents the rainfall in the municipality of origin o during month τ of year t, while $\bar{r}_o$ denotes the historical average precipitation for the same months in that municipality. The index τ covers the six-month growing season (GS). Temperature is calculated in a similar manner, with the average used instead of the sum to generate yearly data. In our main analyses, we focus on data from the Semiarid region's growing season, which spans from November to April. However, our findings remain consistent when we include data from the entire year (see Table G1).

Table G1: **Migration outflows induced by weather shocks (12 months)**

	(1)	(2)	(3)	(4)	(5)	(6)
Rainfall$_{t-1}$	-0.127***	-0.108***	-0.107***	-0.112***	-0.113***	-0.109***
	(0.034)	(0.033)	(0.034)	(0.034)	(0.034)	(0.034)
Rainfall$_{t-2}$			0.027	0.046		
			(0.037)	(0.038)		
Rainfall$_{t-3}$				0.048		
				(0.033)		
Rainfall$_{t}$					-0.015	
					(0.039)	
Rainfall$_{t+1}$						-0.069*
						(0.037)
Observations	14,400	14,400	14,400	14,400	14,400	14,400
Municipalities	960	960	960	960	960	960
R-Squared	0.461	0.465	0.465	0.465	0.465	0.465
Time dummies	✓	✓	✓	✓	✓	✓
Municipality dummies	✓	✓	✓	✓	✓	✓
Temperature shocks	✓	✓	✓	✓	✓	✓
Demographics		✓	✓	✓	✓	✓

Notes: Each observation is a municipality-year cell. The dependent variable is the number of individuals who left the origin municipality divided by the total population in the 1991 Census. Rainfall is measured as log-deviation from the historical average. All specifications include controls for temperature shocks, municipality, and year fixed effects. Columns (2)-(6) control for municipality-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Standard errors are clustered at the grid level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

G.2 Alternative measures of weather

One possible concern regarding our weather measurement is that we focus on rainfall levels, controlling for temperature variation, to predict the flow of migrants leaving the Semiarid region. This approach could be problematic because it does not account for groundwater or other factors that influence water balance. To address this issue, we use the Standardized Precipitation Evapotranspiration Index (SPEI) developed by Vicente-Serrano et al. (2010). The SPEI considers both precipitation and potential evapotranspiration, providing a measure of water balance over a specific period. This index also captures deviations from the historical average (1905-2018) regarding the net water needed for a particular location. An SPEI value of -1 indicates that precipitation is one standard deviation below the historical level required to maintain balance, given potential evapotranspiration. According to Vicente et al. (2010), the SPEI is especially useful for detecting, monitoring, and exploring the consequences of global warming on drought conditions.

We revisit our initial approach to constructing an instrument for in-migration, this time utilizing the Standardized Precipitation-Evapotranspiration Index (SPEI) instead of relying on rainfall and temperature shocks. We calculate the average SPEI for the six-month growing season. As shown in Table G2, this measure can also be used to predict the out-migration rate from municipalities in the Semiarid region. However, the estimates generated are less precise than those presented in Table 4.

We apply the same specification from column 3 of Tables 5 and 6 using this new instrument. The results, displayed in Tables G3 and G4, indicate that our findings are quite similar.

Table G2: **Migration outflows induced by weather shocks: Standardized Precipitation Evapotranspiration Index (SPEI)**

	(1)	(2)	(3)	(4)	(5)	(6)
SPEI$_{t-1}$	-0.033**	-0.041**	-0.038**	-0.042**	-0.045***	-0.043***
	(0.016)	(0.016)	(0.016)	(0.017)	(0.017)	(0.016)
SPEI$_{t-2}$			0.040**	0.041**		
			(0.019)	(0.019)		
SPEI$_{t-3}$				0.032*		
				(0.019)		
SPEI$_{t}$					-0.020	
					(0.016)	
SPEI$_{t+1}$						-0.003
						(0.017)
Observations	14,400	14,400	14,400	14,400	14,400	14,400
Municipalities	960	960	960	960	960	960
R-Squared	0.460	0.465	0.465	0.465	0.465	0.465
Time dummies	✓	✓	✓	✓	✓	✓
Municipality dummies	✓	✓	✓	✓	✓	✓
Temperature shocks	✓	✓	✓	✓	✓	✓
Demographics		✓	✓	✓	✓	✓

Notes: Each observation is a municipality-year cell. The dependent variable is the number of individuals who left the origin municipality divided by the total population in the 1991 Census. All specifications include controls for temperature shocks, municipality, and year fixed effects. Columns (2)-(6) control for municipality-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Standard errors are clustered at the grid level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table G3: **Effects of migration on earnings and employment, using SPEI to predict out-migration**

	Employment			Earnings		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS						
In-migration rate	-0.090	-0.029	-0.061	0.165	0.310	-0.228
	(0.091)	(0.097)	(0.082)	(0.427)	(0.466)	(0.450)
Observations	8,190	8,190	8,190	8,190	8,162	8,179
Municipalities	684	684	684	684	684	684
Panel B: SSIV						
In-migration rate	-0.006	-0.118***	0.112***	-0.925***	-0.638***	-0.789***
	(0.037)	(0.040)	(0.036)	(0.209)	(0.211)	(0.127)
Observations	11,460	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955	955

Notes: This table shows origin-level destination-level OLS and SSIV coefficients of regressions on log earnings and on the employment rate for native workers. Each observation is an origin municipality-year cell. The instrument for the migrant inflow is calculated using the SPEI to predict out-migration distributed by the 1991 share of Semiarid migrants. All regressions include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table G4: **Effects of migration on non-wage benefits, using SPEI to predict out-migration**

	Food	Transport	Health
	(1)	(2)	(3)
Panel A: OLS			
In-migration rate	-0.061	0.027	-0.048
	(0.094)	(0.083)	(0.081)
Observations	8,190	8,190	8,190
Municipalities	684	684	684
Panel B: SSIV			
In-migration rate	-0.244***	-0.118***	-0.115***
	(0.037)	(0.027)	(0.026)
Observations	11,460	11,460	11,460
Municipalities	955	955	955

Notes: This table shows destination-level OLS and origin-level SSIV coefficients of regressions on the proportion of working-age native population receiving non-wage benefits. Each observation is an origin municipality-year cell. The instrument for the migrant inflow is calculated using the SPEI to predict out-migration distributed by the 1991 share of Semiarid migrants. All regressions include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

G.3 Selection into migration

In our analysis, we rely on the hypothesis that, on average, migrants possess lower skill levels than native workers. However, we acknowledge that this is not always the case. As shown in Table 3, the average migrant from the Semiarid region is less educated than the average native worker in the destination municipalities. Additionally, we found that the marginal migrant, who is driven to migrate due to weather shocks, is even less skilled than the average migrant.

To investigate this further, we regress the changes in the proportion of less-educated individuals in the origin municipalities against the average rainfall and temperature shocks. For each decade, we also compute the proportion of years affected by drought, defined as any year when precipitation levels fall below the historical average.

Columns (1)-(3) of Table G5 indicate a positive relationship between average rainfall shocks and the changes in the proportion of less-educated individuals. Moreover, columns (4)-(6) demonstrate that the change in the proportion of less-educated individuals is negatively correlated with the proportion of years that experienced drought within the municipality. To further validate these findings, we incorporate state- and municipality-specific trends.

Given that Brazil has been enhancing access to basic education since the early 1990s, we interpret these results as evidence that migrants who are compelled to move due to weather shocks are, on average, less educated than their counterparts.

Table G5: **Effects of weather shocks on the education level in origin municipalities**

	(1)	(2)	(3)	(4)	(5)	(6)
Average rainfall shock	0.094***	0.196***	0.327***			
	(0.035)	(0.040)	(0.123)			
Proportion of drought years				-0.032***	-0.044***	-0.076***
				(0.012)	(0.014)	(0.025)
Observations	1,920	1,920	1,920	1,920	1,920	1,920
Municipalities	960	960	960	960	960	960
R-Squared	0.358	0.384	0.532	0.358	0.379	0.533
Time dummies	✓	✓	✓	✓	✓	✓
Temperature shocks	✓	✓	✓	✓	✓	✓
Demographics	✓	✓	✓	✓	✓	✓
State dummies		✓		✓		
Municipality dummies			✓			✓

Notes: This table shows the OLS coefficients of regressions of weather shocks on the change in the proportion of less-educated individuals in origin municipalities in the Semiarid region, between 1991 and 2010. Each observation is an origin municipality-decade cell. All regressions include time dummies and control for temperature shocks and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Appendix H Shift-share instrument (SSIV)

In this section, we derive the origin-level SSIV estimator and discuss the identifying assumptions necessary to produce a consistent estimator of the effects of migrant inflows from the Semiarid region on labor markets in the destination municipalities.

We begin with the structural equation (Equation 5). To simplify notation, we omit the time subscript t. According to the Frisch-Waugh-Lovell Theorem, we can re-write the equation as follows:

$$y_d^{\perp} = \beta m_d^{\perp} + \varepsilon_d^{\perp} \tag{H1}$$

where $y_d^{\perp}$ represents the vector of outcomes, while $m_d^{\perp}$ [25] is the observed number of migrants from the Semiarid region who have moved to the destination municipality d. Additionally, $\varepsilon_d^{\perp}$ is the structural residual term. All variables have been residualized to remove the effects of the covariates.

In equation 9, we define the shift-share instrumental variable (SSIV) as follows:

$$\widehat{m}_d = \sum_{o=1}^{O} s_{od} \frac{\widehat{M}_o}{N_d} \tag{H2}$$

where $s_{od} = \dfrac{M_{od}}{\sum_d M_{od}}$ is the share of migrants from origin municipality o who lived in the destination area d in 1991 and $\widehat{M}_o$ is the predicted number of migrants leaving the Semiarid region driven by weather shocks.

The more traditional approach would estimate β using $\widehat{m}_d$ as an instrument for the endogenous migrant inflow $m_d^{\perp}$. In such a case, we would have

$$\hat{\beta} = \frac{\sum_d \widehat{m}_d y_d^{\perp}}{\sum_d \widehat{m}_d m_d^{\perp}} \tag{H3}$$

By the definition of $\widehat{m}_d$ in equation H2 and switching the order of the summation,

$$\hat{\beta} = \frac{\sum_d \left(\sum_o s_{od} \frac{\widehat{M}_o}{N_d} \right) y_d^{\perp}}{\sum_d \left(\sum_o s_{od} \frac{\widehat{M}_o}{N_d} \right) m_d^{\perp}} = \frac{\sum_o \widehat{M}_o \left(\sum_d s_{od} \frac{y_d^{\perp}}{N_d} \right)}{\sum_o \widehat{M}_o \left(\sum_d s_{od} \frac{m_d^{\perp}}{N_d} \right)} = \frac{\sum_o s_o \widehat{M}_o \bar{y}_o}{\sum_o s_o \widehat{M}_o \bar{m}_o} \tag{H4}$$

where $\bar{y}_o = \dfrac{\sum_d s_{od} \frac{y_d^{\perp}}{N_d}}{\sum_d s_{od}}$ is a weighted average of the residualized outcome, normalized by the native population, which uses as weights the destination's average exposure to the shocks $s_o = \sum_d s_{od}$. The same result is valid for the endogenous variable $m_d^{\perp}$, meaning that we can estimate the following IV regression at the origin municipality level:

$$\bar{y}_o = \beta \bar{m}_o + \bar{\varepsilon}_o \tag{H5}$$

[25] To aid in the interpretation of the coefficients, we normalize this measure by dividing it by the working-age native population in 1991. Therefore, $m_d = \frac{M_d}{N_d}$

using the predicted number of migrants from the Semiarid region, $\widehat{M_o}$, as instrumental variable and weighting by the average exposure s_o.

This derivation is almost identical to that presented by Borusyak et al. (2021), except that we must divide both variables by the predetermined native population. Their equivalence result shows that the parameter β can be estimated at the level of the identifying variation, which in our case is the origin municipality hit by weather shocks.

Table H1: **SSIV First Stage**

	(1)	(2)	(3)
Panel A: OLS			
mnrate	0.911***	0.910***	0.929***
	(0.016)	(0.016)	(0.020)
F-statistic	3426.56	3428.53	2079.10
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Effective sample size	7,299	7,299	7,299
Time dummies		✓	✓
Baseline			✓

Notes: This table shows the SSIV first-stage coefficients of the origin-level weighted average of the endogenous inflow of migrants at the destinations against the predicted number of migrants from the Semiarid region. Each observation is an origin municipality-year cell. The F-statistic is calculated as the square of the coefficient t-statistic (see Borusyak et al., 2021). The effective sample size is the inverse of the HHI of the origin-level exposure. Column (2) includes time dummies while Column (3) also controls for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Regressions are weighted by the working-age native population in 1991. Standard errors cluster by the municipality of origin in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Appendix I Spatial correlation in weather shocks

Weather events are likely to be correlated across different geographic areas. Figure 1 illustrates that precipitation levels in the Semiarid region are similar among neighboring municipalities. This correlation could undermine the validity of our estimator, as discussed in *Assumption 2 (Many uncorrelated shocks)* in Appendix H.

To address this concern, we re-construct our instrument by aggregating the regions from which migrants originate, using either microregions or mesoregions instead of municipalities. According to IBGE (1990), microregions are defined as "groups of economically integrated municipalities that share borders and similar production structures." Mesoregions consist of collections of microregions, and not all municipalities within these areas share borders.[26] In Brazil, there are 5,565 municipalities, 361 microregions, and 87 mesoregions overall. Within the Semiarid, there are 960 municipalities, 137 microregions, and 35 mesoregions.

The rationale behind this approach is that while weather shocks may be spatially correlated among adjacent municipalities, such correlation is expected to diminish as we consider larger geographic areas. Table I2 presents Moran's index of spatial correlation for rainfall shocks across the three geographic aggregates in columns 1-3.[27] As expected, neighboring municipalities display a correlation above 0,24, but it decreases rapidly as we aggregate up to micro and meso regions, to 0,16 and 0,07, respectively.

Table I2 illustrates the relationship between rainfall shocks and migration outflows. Column 1 mirrors the information presented in Table 4 for reference. Columns 2 and 3 display nearly identical point estimates and precision, suggesting that aggregating origin areas does not result in a significant loss of information. Next, we estimate our primary specification from Column (3) in Tables 5 and 6 using instruments that correspond to micro and mesoregion-level aggregation. Tables I3 to I4 demonstrate that our findings linking migration with earnings, employment, and non-wage benefits are very similar to those estimated at the municipality level. However, standard errors increase significantly, as expected, due to the reduced number of units available for variation analysis. Overall, the results indicate that spatial correlation among rainfall shocks in origin municipalities does not introduce relevant bias in our study.

[26] Table I1 provides summary statistics of our main variables for both levels of aggregation.

[27] Moran's I is calculated using the following formula:

$$I = \frac{1}{\sum_i \sum_j w_{ij}} \times \frac{\sum_i \sum_j w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\frac{1}{N}(y_i - \bar{y})^2} \tag{I1}$$

Essentially, it is a correlation coefficient weighted by an appropriate matrix that models how different units are related across space. We use a contiguity matrix with the queen criterion, meaning that two localities i and j sharing either borders or vertices are considered 'neighbors', and the entry w_{ij} has a positive value. Non-adjacent pairs receive a zero weight. As discussed by Beenstock et al. (2019), Moran's I can be calculated for each period and averaged out with panel data.

Table I1: **Summary statistics: Micro- and meso-regions in the Semiarid**

Panel A: Origin (Semiarid)	Mean	Std. Dev.	Min	Max	Obs
Rainfall shock	-0.01	0.20	-0.68	0.46	2,055
Temperature shock	0.01	0.01	-0.01	0.05	2,055
Out-migration	1,501.39	1,370.34	6.00	9,685.00	2,055
Out-migration rate (p.p.)	1.08	0.41	0.12	3.16	2,055
Population	149,866	128,901	4,968	752,719	2,055
Area	8,299.88	8,886.98	278.43	55,360.33	2,055
Number of municipalities	8.20	4.56	2.00	26.00	2,055

Panel B: Destination (Non-Semiarid)	Mean	Std. Dev.	Min	Max	Obs
Rainfall shock	-0.02	0.19	-0.68	0.44	525
Temperature shock	0.01	0.01	-0.01	0.04	525
Out-migration	5,876.85	5,763.83	51.00	34,800.00	525
Out-migration rate (p.p.)	1.09	0.38	0.24	3.16	525
Population	586,618	526,749	15,499	2,349,152	525
Area	37,854.55	39,127.42	3,767.55	128,389.96	525
Number of municipalities	37.20	21.51	10.00	118.00	525

Notes: Rainfall is measured in mm. Temperature is measured in degrees Celsius. Migration outflow (inflow) rate is the share of migrants over the local (native) population. The area is measured in km^2.

Table I2: **Migration outflows induced by weather shocks according to different aggregation levels**

	(1)	(2)	(3)
	Municipality	Micro-region	Meso-region
Rainfall$_{t-1}$	-0.094***	-0.091***	-0.091***
	(0.028)	(0.030)	(0.023)
Observations	14,400	2,055	525
Regions	960	137	35
R squared	0.465	0.772	0.869
Moran's I	0.235	0.158	0.075

Notes: Each observation is a region-year cell. The dependent variable is the number of individuals who left the origin region divided by the total population in the 1991 Census. Rainfall is measured as log-deviation from the historical average. All specifications control for temperature shocks and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies, and include municipality, and year fixed effects. Moran's I show the spatial correlation in rainfall shocks among origin regions. Standard errors are clustered at the respective region level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table I3: **Effects of migration on earnings according to different aggregation levels**

	Employment			Earnings		
	Overall	Formal	Informal	Overall	Formal	Informal
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Municipality						
In-migration rate	-0.014	-0.127***	0.113***	-0.954***	-0.672***	-0.808***
	(0.036)	(0.040)	(0.036)	(0.209)	(0.212)	(0.129)
Observations	11,460	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955	955
Panel B: Micro-region						
In-migration rate	0.002	-0.117**	0.119**	-0.926***	-0.631**	-0.807***
	(0.063)	(0.059)	(0.060)	(0.319)	(0.309)	(0.210)
Observations	1,644	1,644	1,644	1,644	1,644	1,644
Municipalities	137	137	137	137	137	137
Panel C: Meso-region						
In-migration rate	0.016	-0.123	0.140	-0.945	-0.615	-0.830**
	(0.098)	(0.106)	(0.102)	(0.580)	(0.560)	(0.353)
Observations	420	420	420	420	420	420
Municipalities	35	35	35	35	35	35

Notes: This table shows origin-level SSIV coefficients on changes in log earnings by sector, for different levels of aggregation. Each observation is an origin region-year cell. The informal sector also includes self-employed workers. All specifications include time and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Panel A replicates the same results from Table 5. In Panels B and C, we aggregate the origin-level shocks at the micro- and meso-region levels, respectively. All regressions are weighted by native working-age population in 1991. Standard errors clustered at the respective aggregation level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table I4: **Effects of migration on employment according to different aggregation levels**

	Food	Transport	Health
	(1)	(2)	(3)
Panel A: Municipality			
In-migration rate	-0.236***	-0.113***	-0.111***
	(0.037)	(0.026)	(0.026)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Panel B: Micro-region			
In-migration rate	-0.224***	-0.088**	-0.102***
	(0.058)	(0.045)	(0.038)
Observations	1,644	1,644	1,644
Municipalities	137	137	137
Panel C: Meso-region			
In-migration rate	-0.238**	-0.087	-0.104
	(0.095)	(0.072)	(0.065)
Observations	420	420	420
Municipalities	35	35	35

Notes: This table shows origin level SSIV coefficients of change in the proportions of working-age native population who receive health insurance, food, or transport subsidies. Each observation is an origin region-year cell. The informal sector also includes self-employed workers. All specifications include time and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water). Panel A replicates the same results from Table 6. In Panels B and C, we aggregate the origin-level shocks at the micro- and meso-region levels, respectively. All specifications use the same set of controls defined in Table 6. All regressions are weighted by native working-age population in 1991. Standard errors clustered at the respective aggregation level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.